

第三十四章 PCIe 测试

实验 Vivado 工程为 “PCIe_test”。

本实验使用 Xilinx 提供的 PCIe XDMA 来完成一个简单的 PCIe，读写实验，来测试 PCIe 的速度。PCIe 2.0，每个 lane 的速度是 5Gbit/s，由于编码、协议等开销，可用带宽可以达到 80%。由于 ZYNQ 的 HP 只有 64 位，如果速度跑到 150MHz，带宽只有 9.6Gbit/s，不能够满足 PCIe 2.0x8 的带宽要求，因此选用 PL 端的 DDR3 做数据缓存。

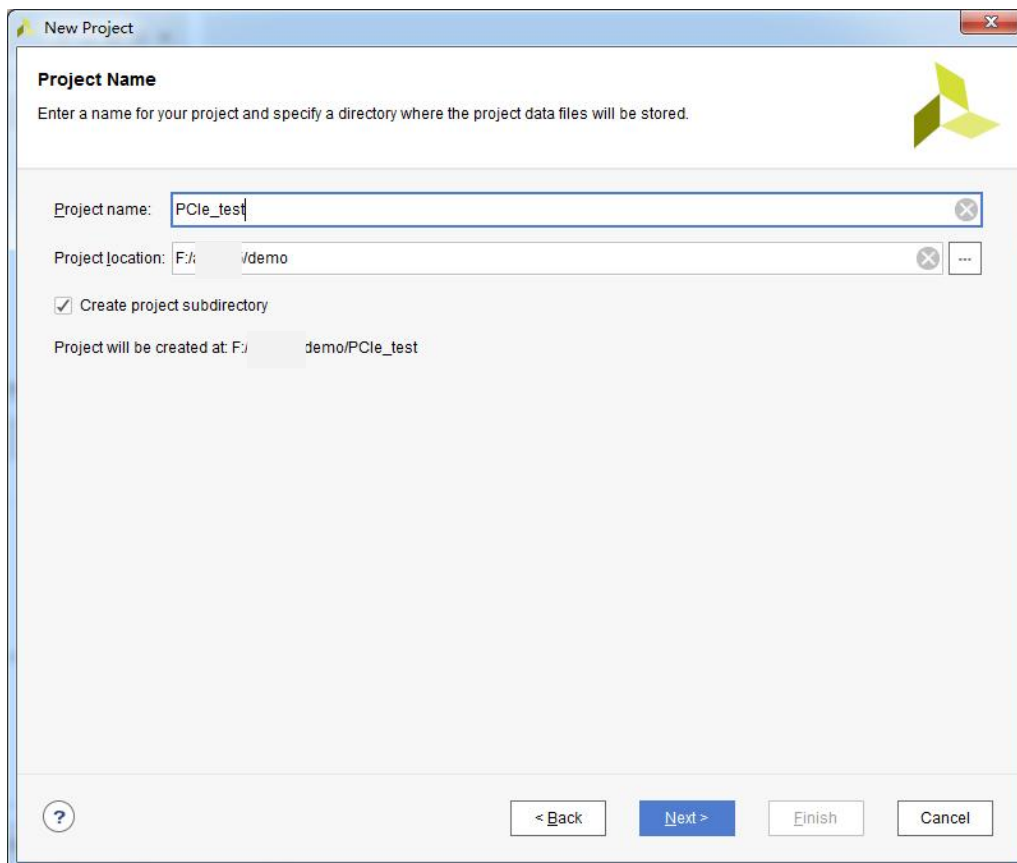
开发板提供 Windows 7 64 位和 Windows 10 64 位驱动程序，已经驱动程序的源代码，驱动程序使用 VS2015 开发，开发板也提供了 Linux 版本的驱动源码和测试程序。

为了在 Windows 下更直观的测试，芯驿电子使用 QT 开发了一些带界面的测试程序，方便测试。

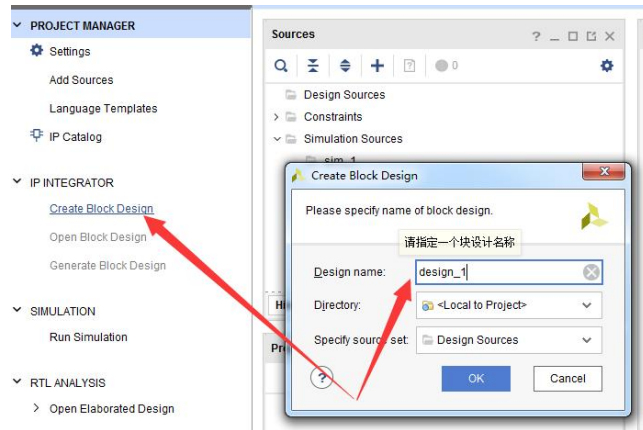
34.1 Vivado 工程建立

34.1.1 ZYNQ 配置

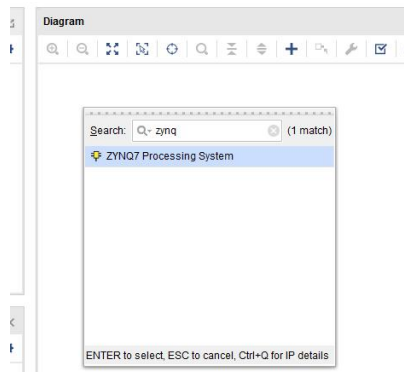
- 1) 新建一个名为 PCIe_test 的 Vivado 工程



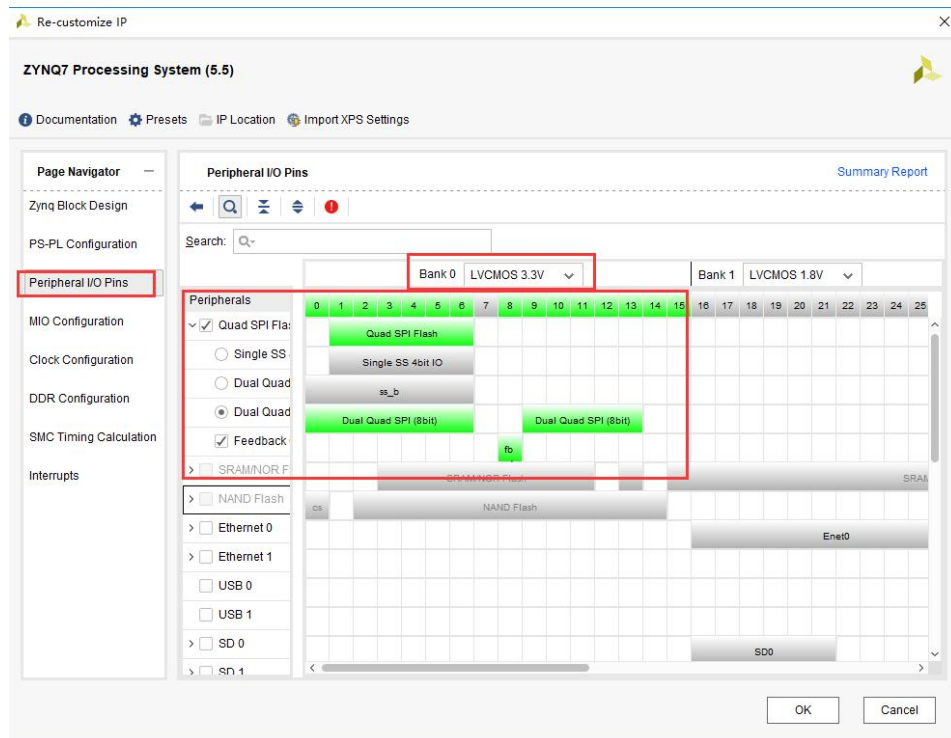
2) 新建一个 Block 设计



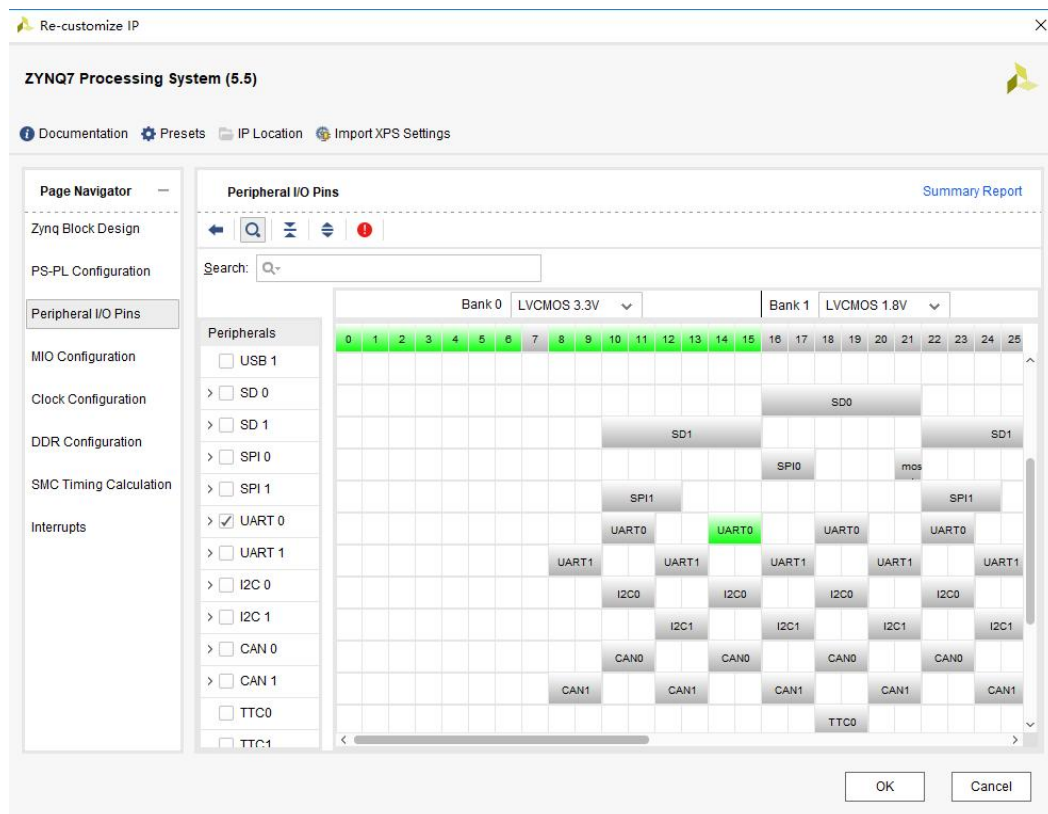
3) 添加 ZYNQ 处理器



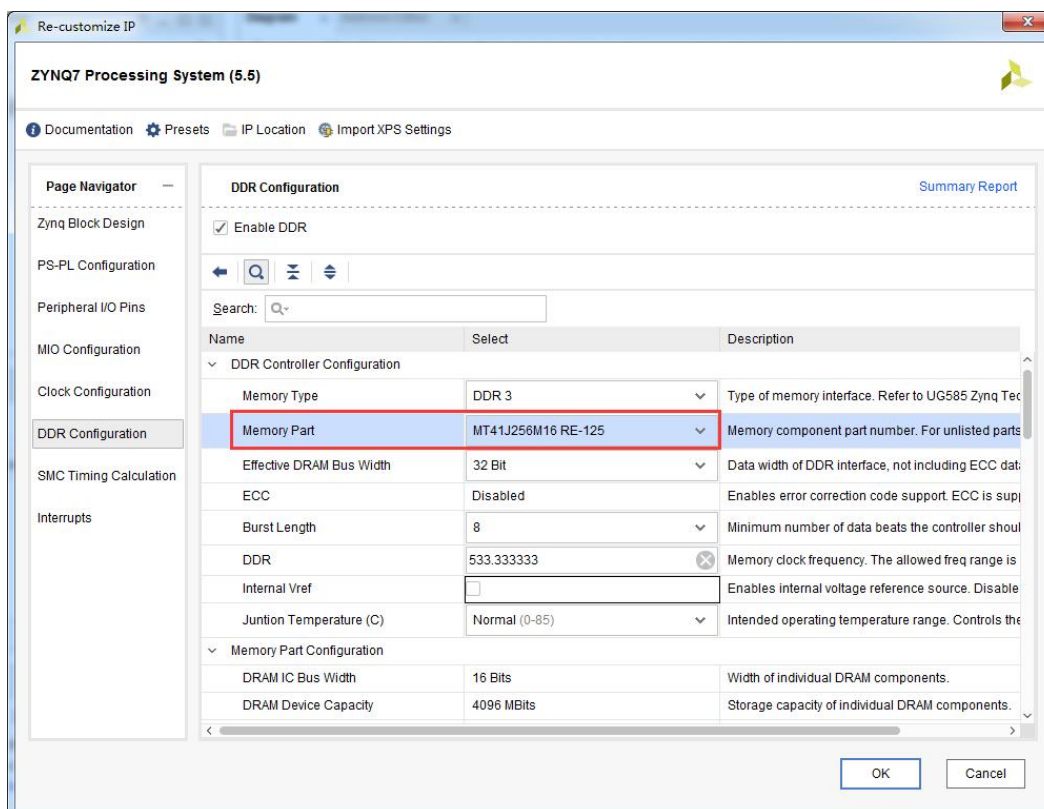
4) 配置 MIO, BNAK 0 电平为 LVCMOS 3.3V, Bnak1 电平为 LVCMOS 1.8V, 使能 QSPI, 这样才能固化程序, PCIe 才能正常使用。



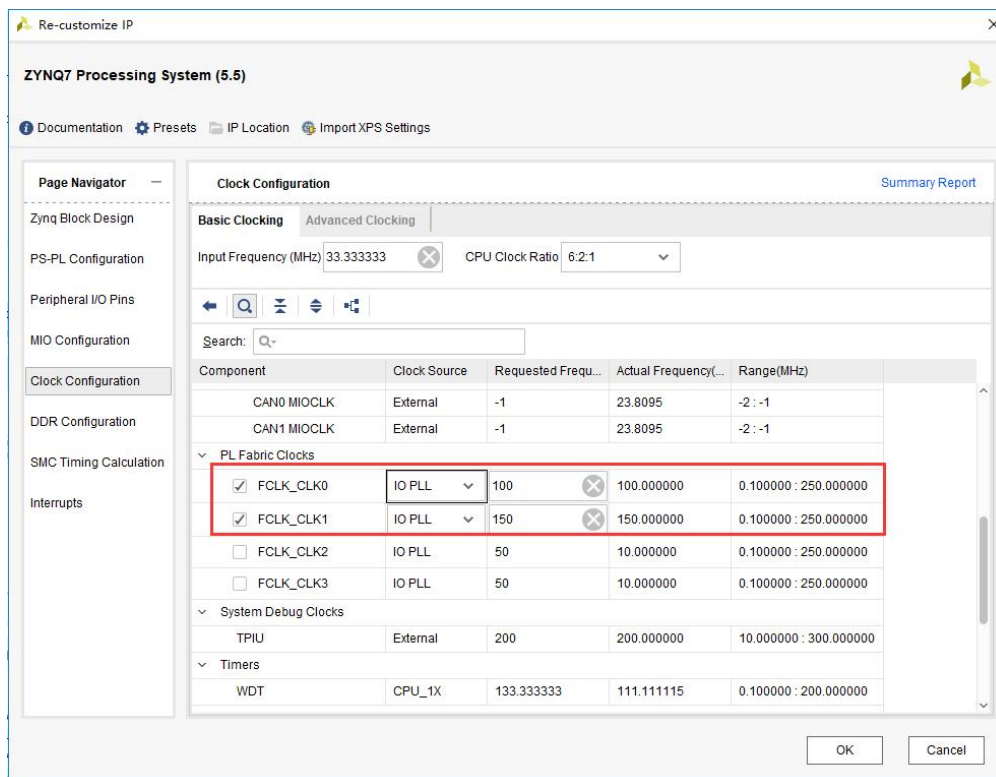
5) 为了调试方便, 使能 UART0



6) ddr3 配置, ddr3 选择 MT41J256M16 RE-125

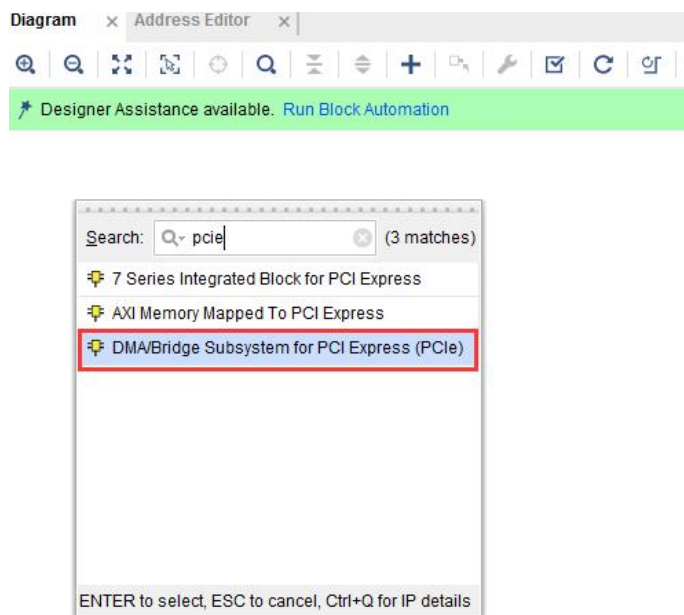


- 7) 配置 FCLK_CLK0 为 100MHz 用于 GP0 时钟, FCLK_CLK1 为 150MHz 用于 axi_interconnect, 点击 OK.

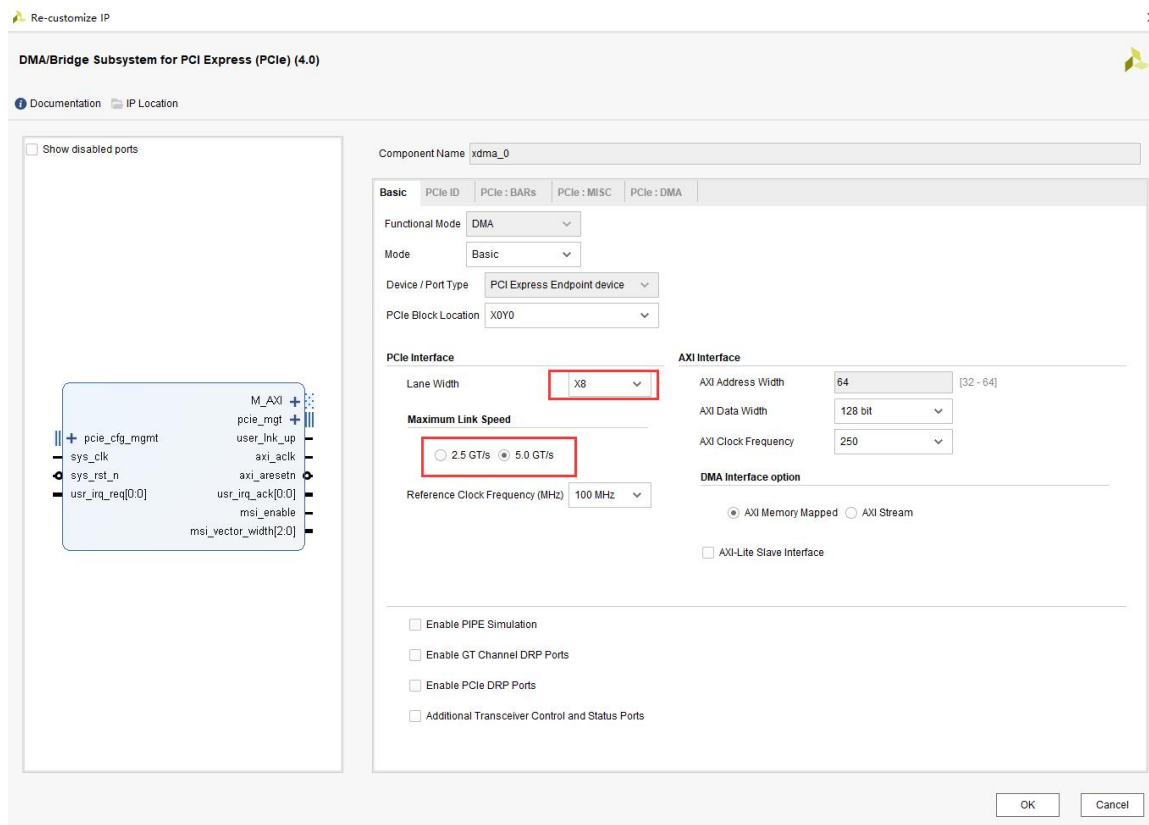


34.1.2 PCIe xdma 配置

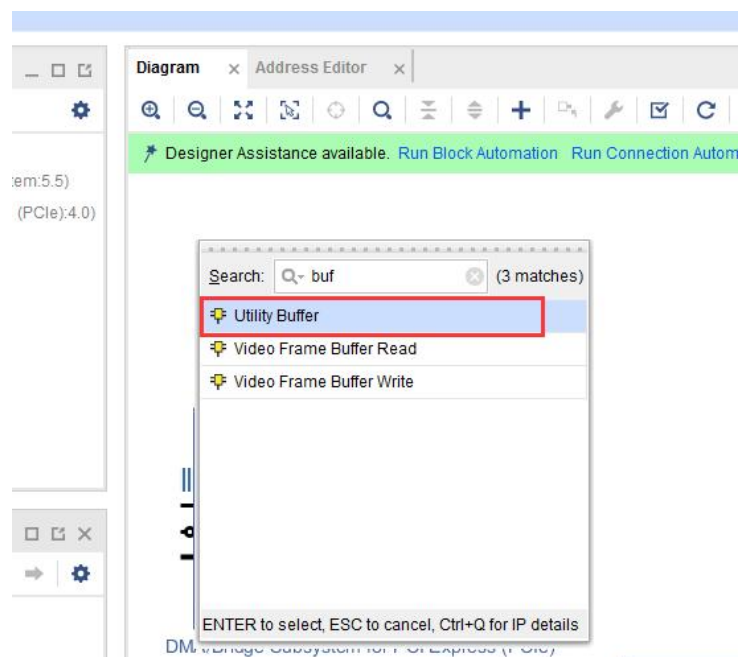
8) 添加 PCIe DMA 子系统



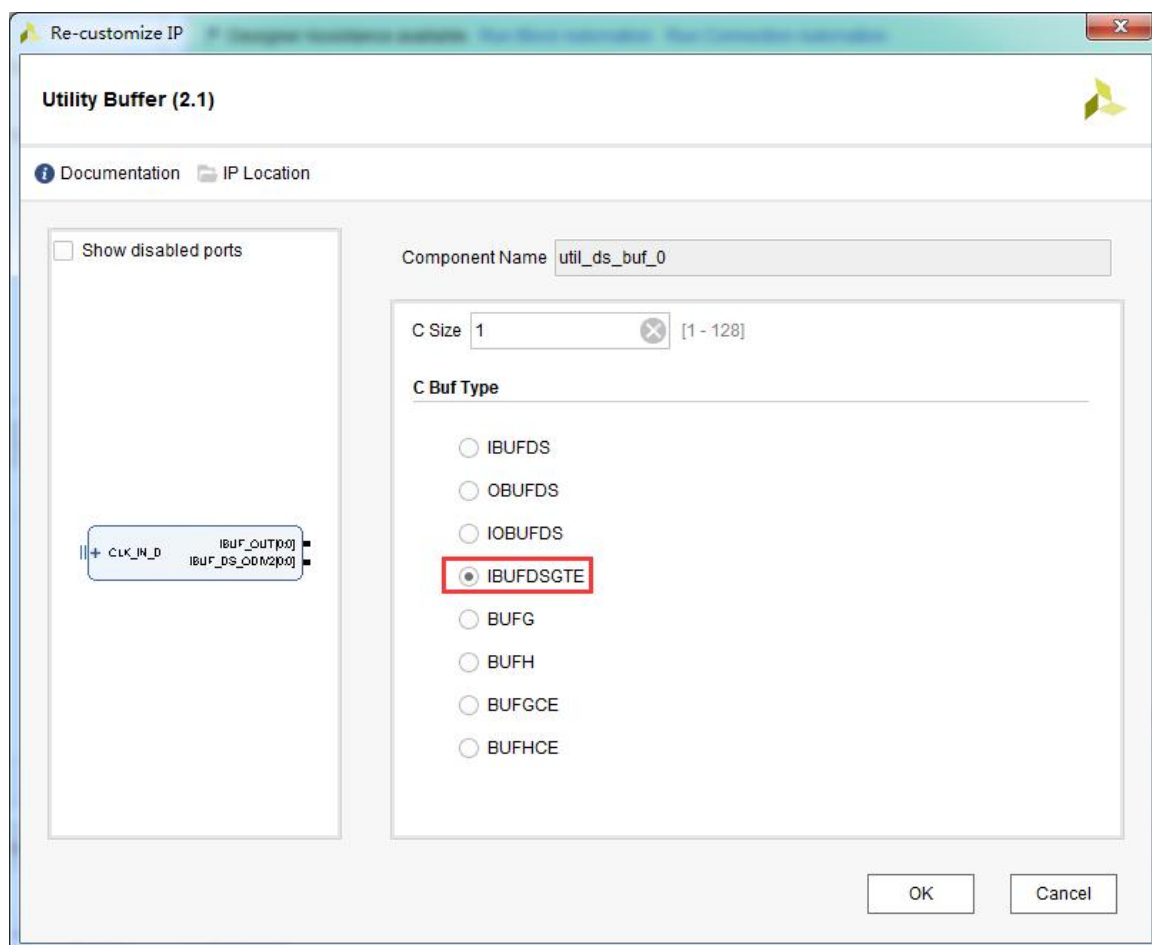
9) 配置 PCIe xdma 参数，Lane 宽度选择 X8，速率选择 5.0 GT/s，其他参数都保持默认。



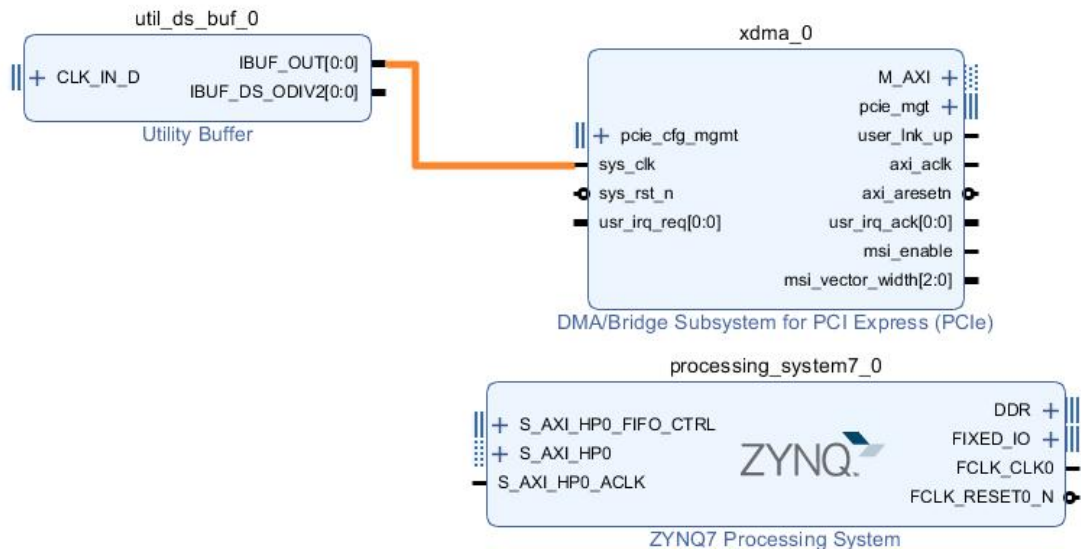
10) 添加 PCIe 差分参考时钟缓冲



11) 配置为收发器时钟差分缓冲

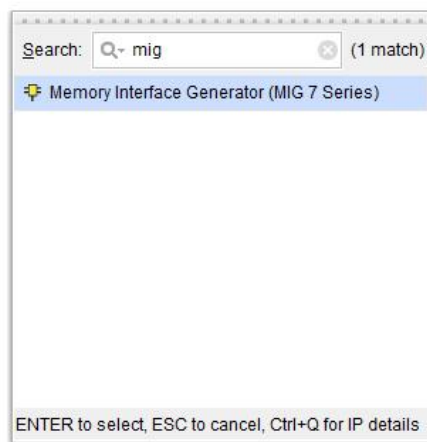


12) 连接 PCIe 参考时钟

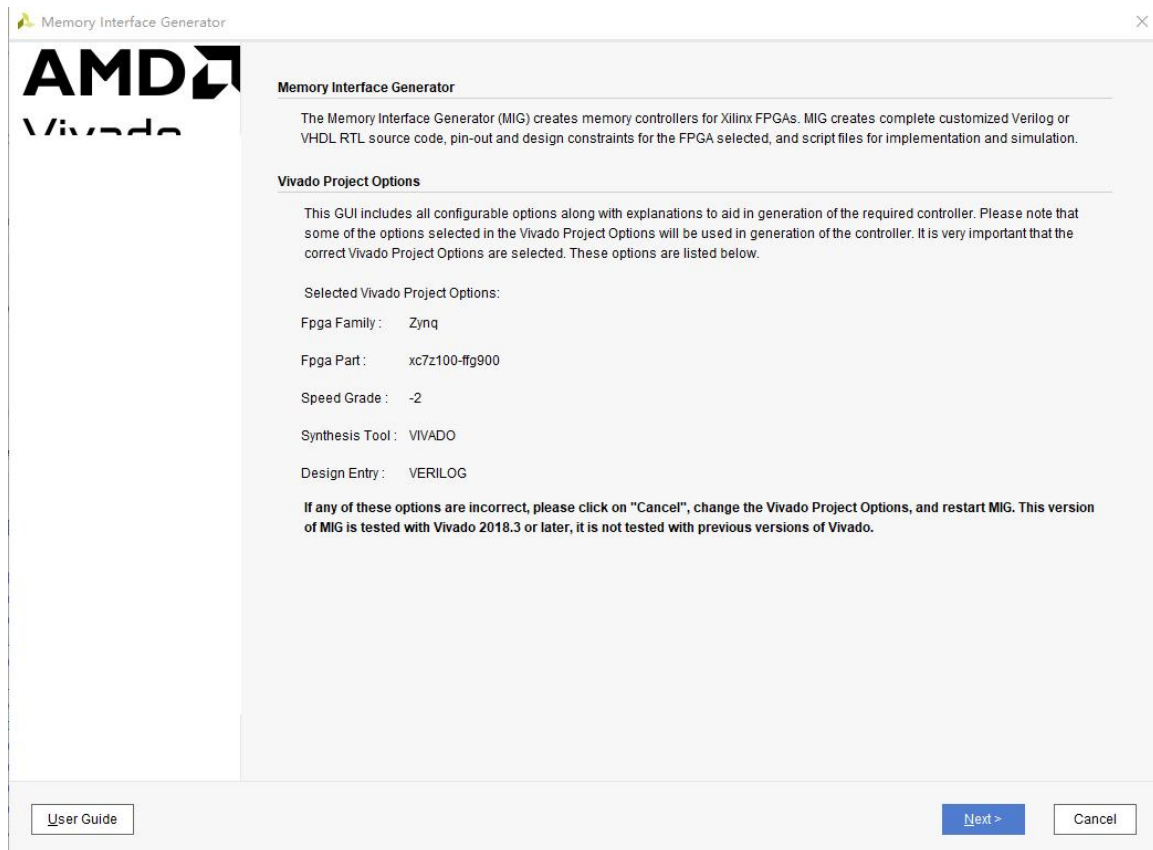


34.1.3 PL 端 DDR3 配置

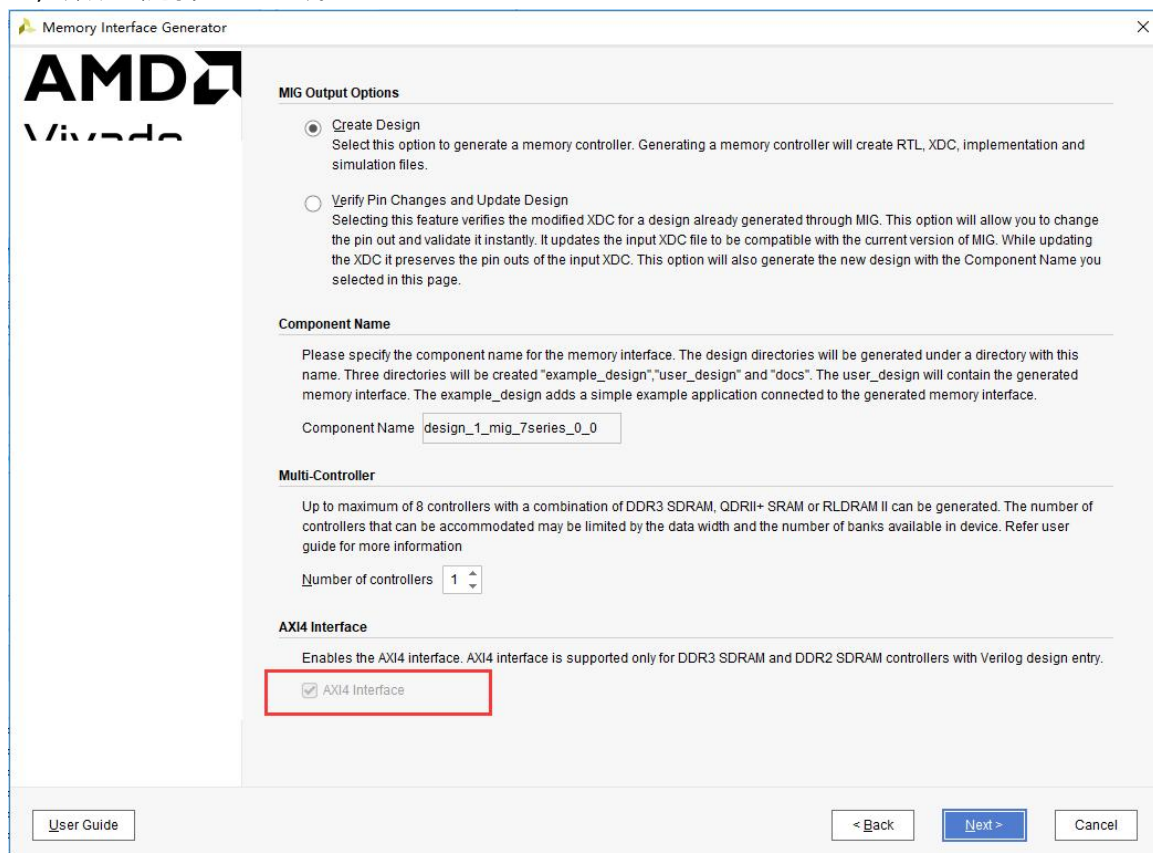
13) 添加 PL 端 DDR3 控制器 MIG



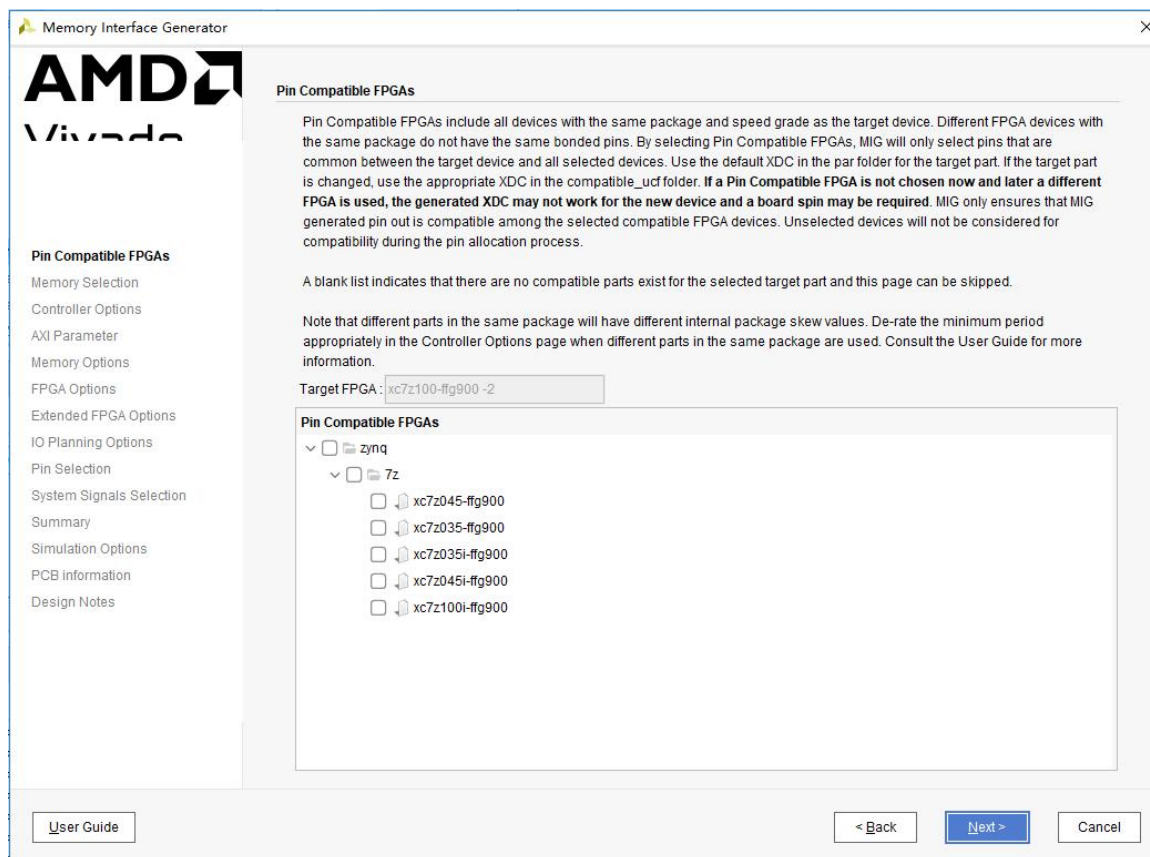
14) 与 PL 端读取 DDR3 配置类似



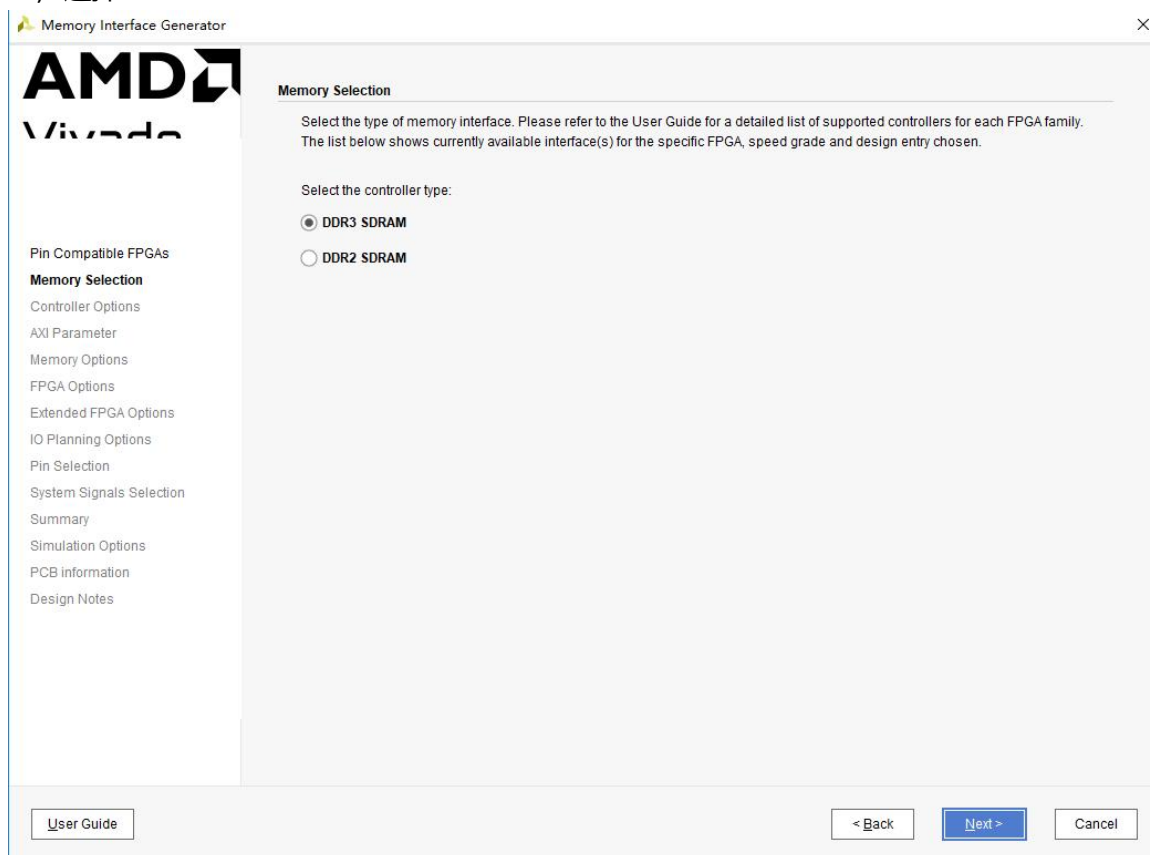
15) 默认会选择 AXI4 总线



16) 点击下一步



17) 选择 DDR3



18) 选择型号 MT41K256M16XX-125

Memory Interface Generator

Options for Controller 0 - DDR3 SDRAM

Clock Period: Choose the clock period for the desired frequency. The allowed period range(1072 - 3300) is a function of the selected FPGA part and FPGA speed grade. Refer to the User Guide for more information. 1,250 ps 800.0 MHz

PHY to Controller Clock Ratio: Select the PHY to Memory Controller clock ratio. The PHY operates at the Memory Clock Period chosen above. The controller operates at either 1/4 or 1/2 of the PHY rate. The selected Memory Clock Period will limit the choices. 4:1

Vccaux_io: Vccaux_io must be set to 2.0V in the High Performance banks for the highest data rates. Vccaux_io is not available in the High Range banks. Note that Vccaux_io is common to groups of banks. Consult the 7 Series Datasheets and FPGA SelectIO Resources User Guide for more information. 2.0V

Memory Type: Select the memory type. Type(s) marked with a warning symbol are not compatible with the frequency selection above. Components

Memory Part: Select the memory part. Part(s) marked with a warning symbol are not compatible with the frequency selection above. Find an equivalent part or create a part using the "Create Custom Part" button if the part needed is not listed here. The "Create Custom Part" feature is not supported for RLD RAM II. MT41K256M16XX-125 Create Custom Part

Memory Voltage: Select the Voltage of the Memory part selected. 1.5V

Data Width: Select the Data Width. Parts marked with a warning symbol are not compatible with the frequency and memory part selected above. 32

ECC: MIG supports ECC for 72 bit data width configuration. To be able to select ECC, select a data width Disabled

Memory Details: 4Gb, x16, row:15, col:10, bank:3, data bits per strobe:8, with data mask, single rank, 1.35V, 1.5V

User Guide < Back Next > Cancel

19) AXI 数据宽度选择 256

Memory Interface Generator

Axi Parameter Options C0 - DDR3 SDRAM

Data Width

AXI DATA WIDTH: Data width of AXI read & write channels. The data width is less than or equal to user interface data width with the possible values 32, 64, 128, 256 & 512. 256

Arbitration Scheme

Select the arbitration scheme between the read and write address channels RD_PRI_REG

Narrow Burst Support

Enables logic to support narrow bursts on the AXI4 slave interface. Can be set to zero if no masters in the system issue narrow bursts and all the data widths are equal. (1-Enable, 0-Disable) 0

Address Width

AXI4 address width of read and write address channels. 31

ID Width

AXI4 ID width for read and write channels. AXI4 ID is used as the identification tag for write or read address group of signals 1

User Guide < Back Next > Cancel

20) 输入时钟选择 200MHz

Memory Interface Generator

AMD Vivado

Pin Compatible FPGAs
Memory Selection
Controller Options
AXI Parameter
Memory Options
FPGA Options
Extended FPGA Options
IO Planning Options
Pin Selection
System Signals Selection
Summary
Simulation Options
PCB Information
Design Notes

Memory Options C0 - DDR3 SDRAM

Input Clock Period: Select the period for the PLL input clock (CLKIN). MIG determines the allowable input clock periods based on the Memory Clock Period entered above and the clocking guidelines listed in the User Guide. The generated design will use the selected Input Clock and Memory Clock Periods to generate the required PLL parameters. If the required input clock period is not available, the Memory Clock Period must be modified.

5000 ps (200 MHz)

☐ Select Additional Clocks (if required)

MIG can generate up to 5 additional clocks to be used in Fabric logic. This will be generated from the same MMCM which is used for generation of UI_CLK. The first clock (Clock 0) has a wider range of choices. All the values in the additional clocks drop downs are calculated considering the MMCM VCO frequency as 1250 ps (800 MHz) Mhz. For complete details on clocking of MIG, refer to MIG User Guide.

Clock	Selection	D
Clock 0	NONE	D = 1
Clock 1	NONE	D = 1
Clock 2	NONE	D = 1
Clock 3	NONE	D = 1
Clock 4	NONE	D = 1

Choose the Memory Options for the memory device. Memory Option selections are restricted to those supported by the controller. Consult the memory vendor data sheet for more information.

Read Burst Type and Length

The burst type determines the data ordering within a burst. Consult the memory datasheet for more information. Burst length 8 is the only supported value.

Sequential

User Guide < Back Next > Cancel

21) 系统时钟选择差分，参考时钟选择系统时钟，系统复位选择低电平有效

Memory Interface Generator

AMD Vivado

Pin Compatible FPGAs
Memory Selection
Controller Options
AXI Parameter
Memory Options
FPGA Options
Extended FPGA Options
IO Planning Options
Pin Selection
System Signals Selection
Summary
Simulation Options
PCB Information
Design Notes

System Clock

Choose the desired input clock configuration. Design clock can be Differential or Single-Ended.

System Clock: Differential

Reference Clock

Choose the desired reference clock configuration. Reference clock can be Differential or Single-Ended.

Reference Clock: Use System Clock

System Reset Polarity

Choose the desired System Reset Polarity.

System Reset Polarity: ACTIVE LOW

Internal Vref

Internal Vref can be used to allow the use of the Vref pins as normal IO pins. This option can only be used at 800 Mbps and lower data rates. This can free 2 pins per bank where inputs are used. This setting has no effect on banks with only outputs.

Internal Vref: ☐

IO Power Reduction

Significantly reduces average IO power by automatically disabling DQ/DQS I/Os and internal terminations during WRITES and periods of inactivity.

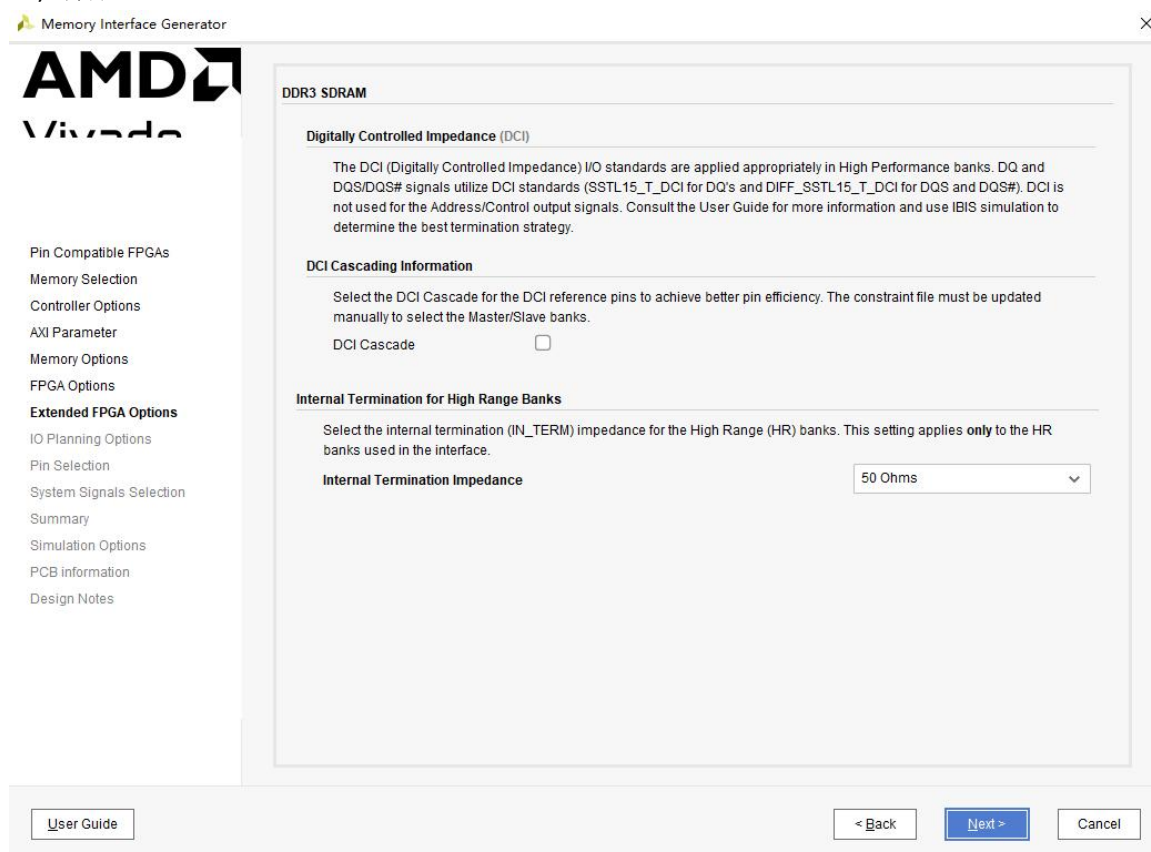
IO Power Reduction: ON

XADC Instantiation

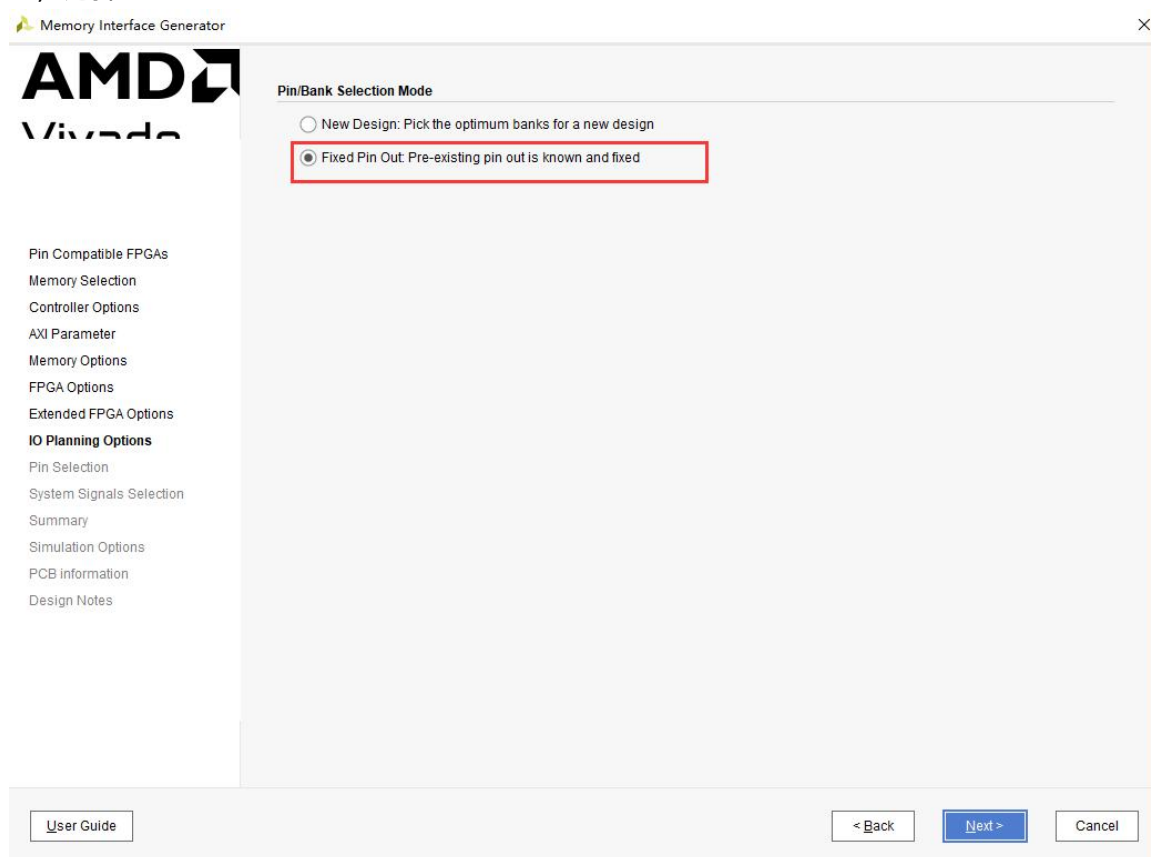
The memory interface uses the temperature reading from the XADC block to perform temperature compensation and keep the read DQS centered in the data window. There is one XADC block per device. If the XADC is not currently used anywhere in the design, enable this option to have the block instantiated. If the XADC is already used, disable this MIG option. The user is then required to provide the temperature value to the top level 12-bit device_temp_i input port. Refer to Answer Record 51687 or the

User Guide < Back Next > Cancel

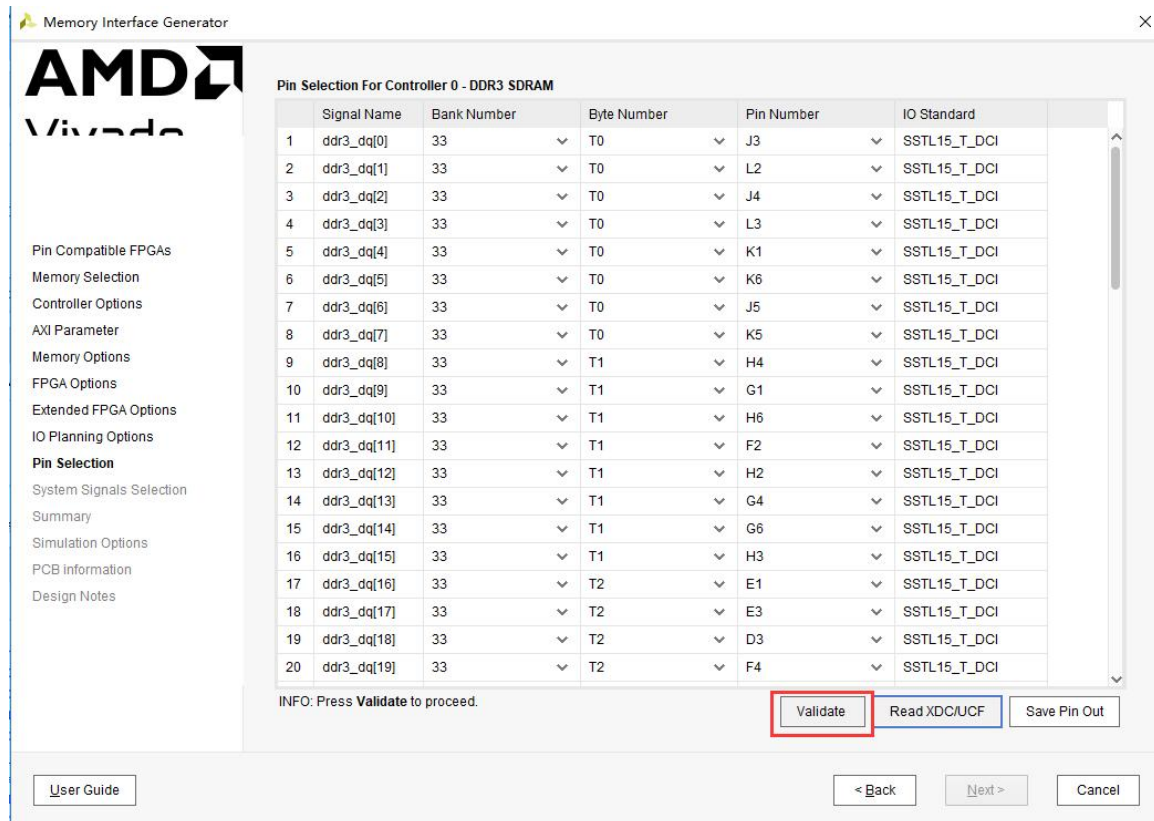
22) 默认



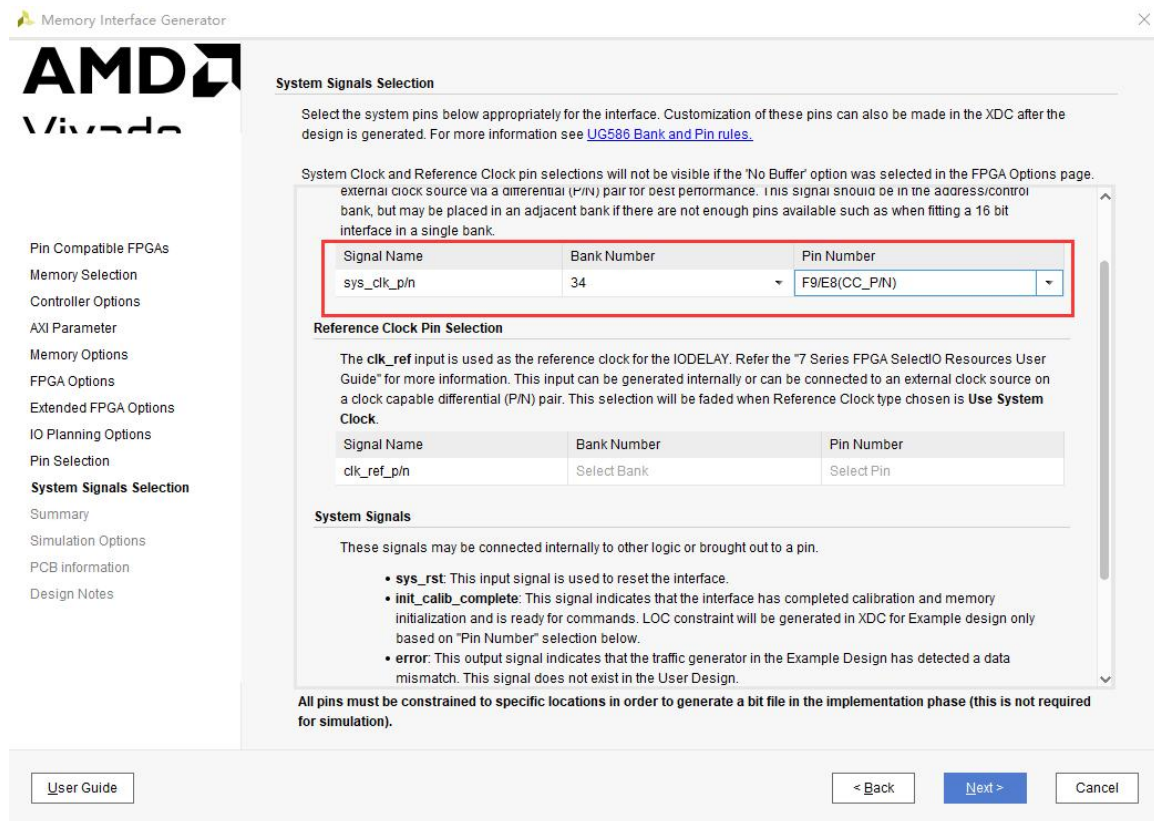
23) 选择 Fixed Pin Out



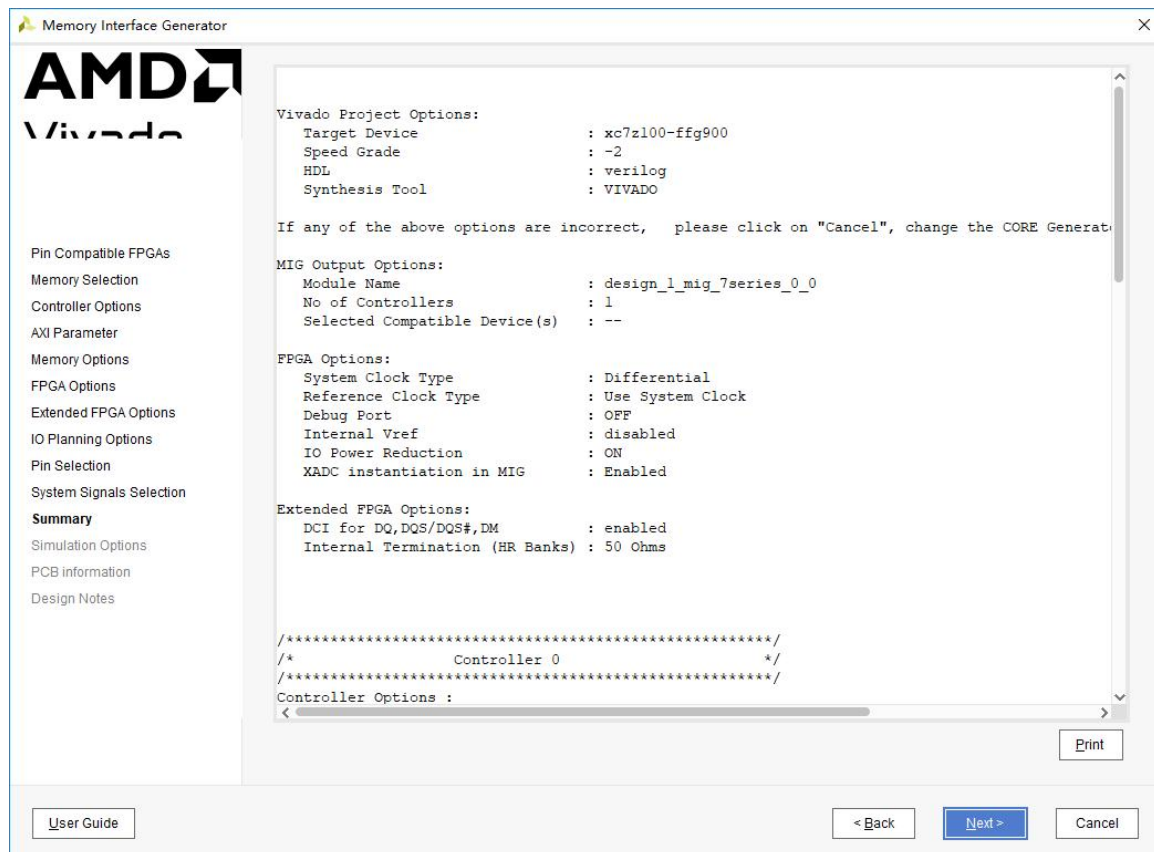
24) 与 PL 端读取 DDR3 一样，读取 UCF 文件（在工程目录下），并且点击 Validate 校验。



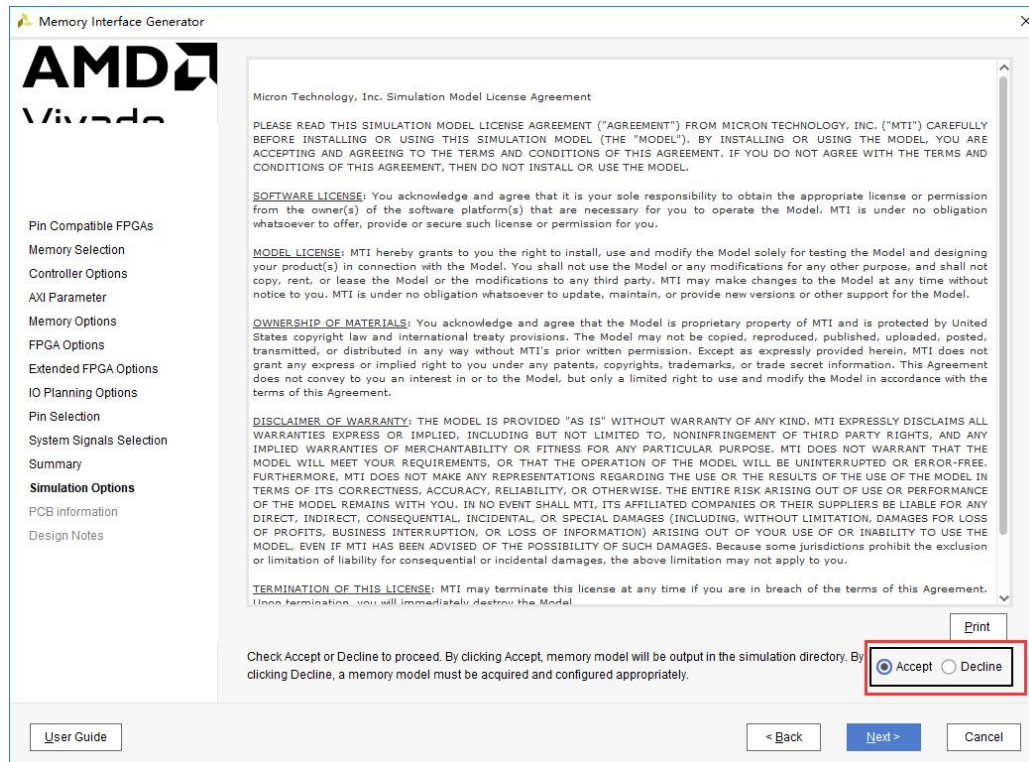
25) 通过原理图可知，系统时钟选择 F9/E8 引脚



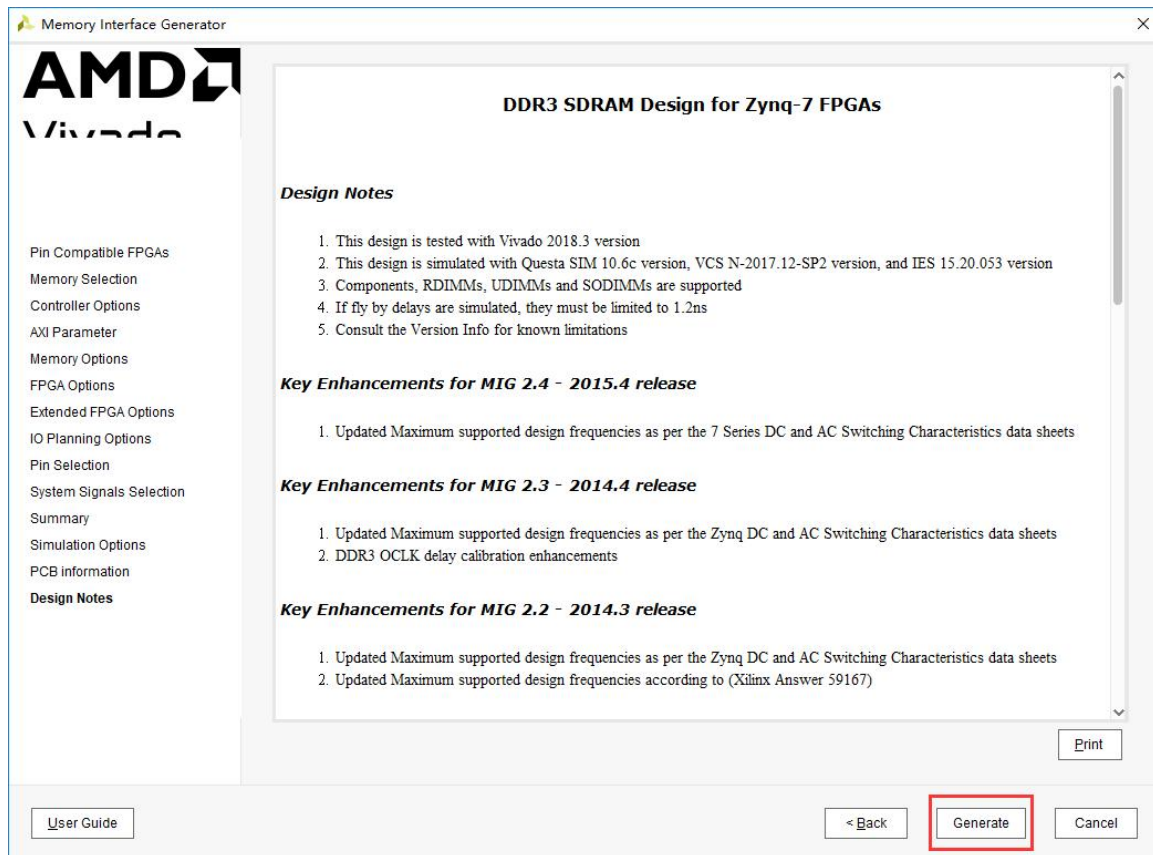
26) 默认



27) 选择 Accept

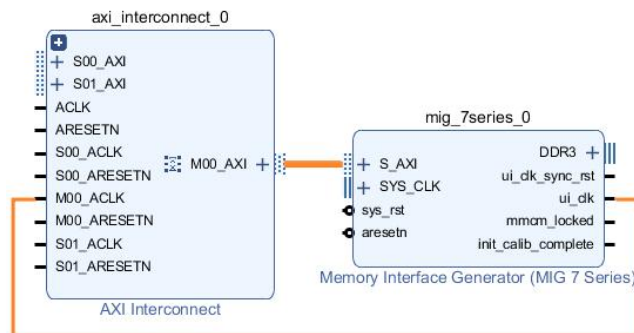


28) 最后选择 Generate

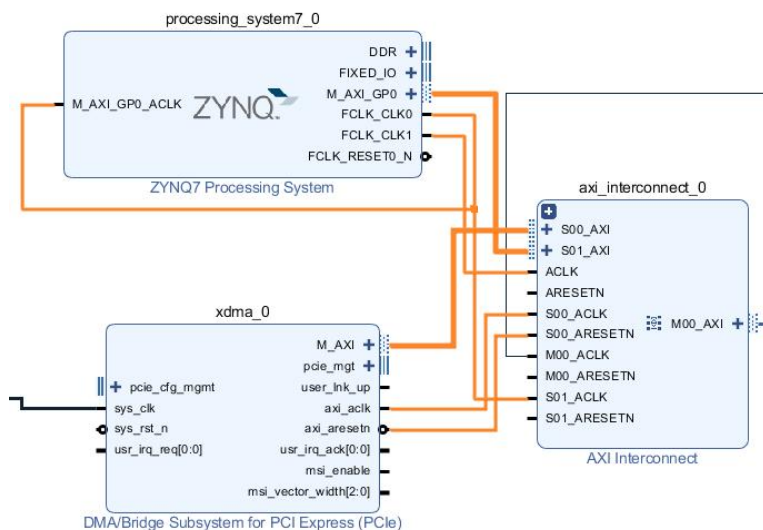


34.1.4 模块连接

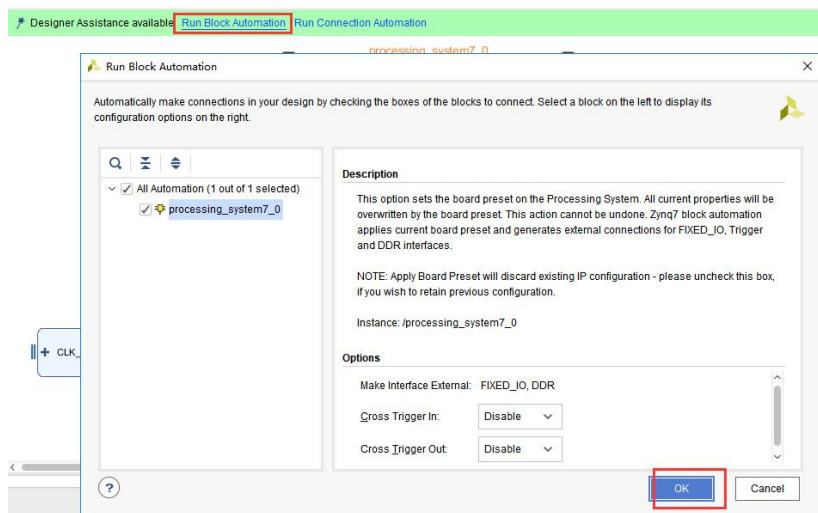
29) 添加 axi_interconnect, 连接 M00_AXI 到 DDR3 控制器的 S_AXI, ui_clk 连接到 M00_ACLK



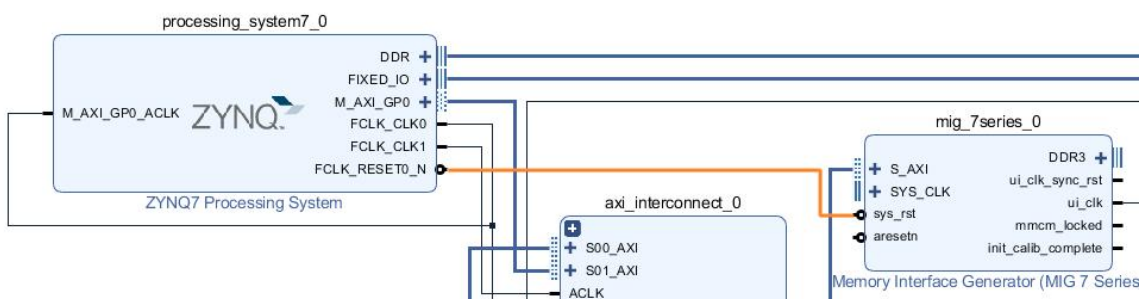
30) 连接处理器 GP0 和 xdma 的 M_AXI 到 axi_interconnect, 连接 FCLK_CLK0 到 S01_ACLK, 连接 FCLK_CLK1 到 ACLK, 用于 interconnect 异步时钟。xdma 的 axi_aclk 连接到 S00_ACLK, xdma 的 axi_aresetn 连接到 S00_ARESETN。



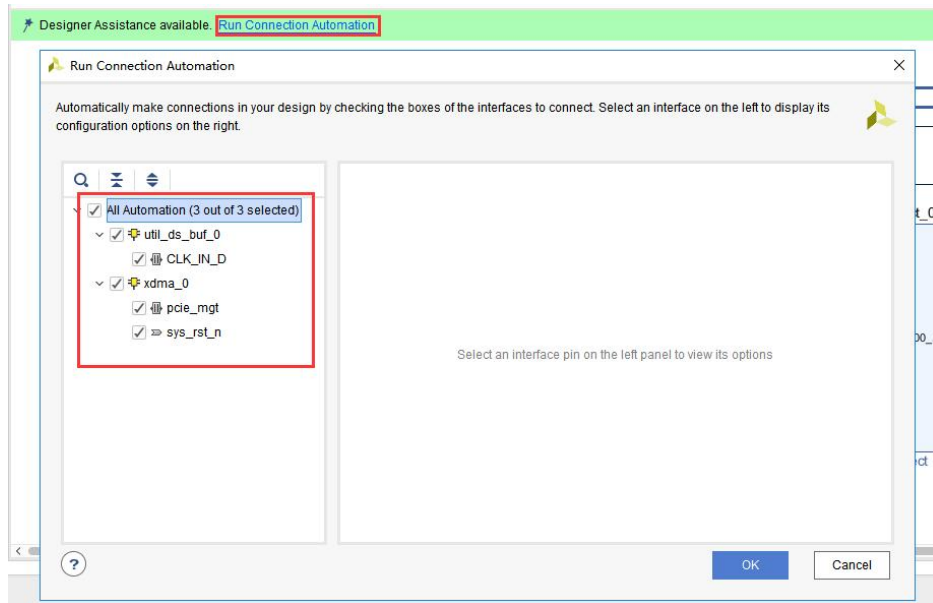
31) 点击 Run Block Automation



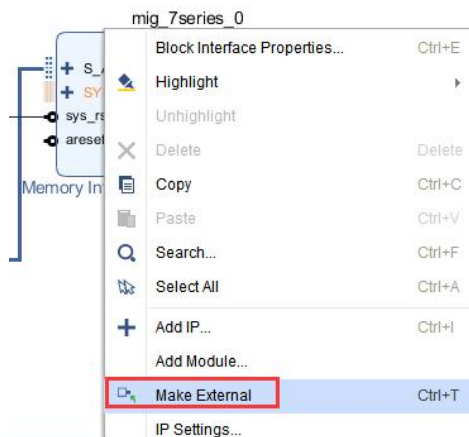
32) 连接 MIG 核的 sys_rst 到 ZYNQ 核的 FCLK_RESET0_N



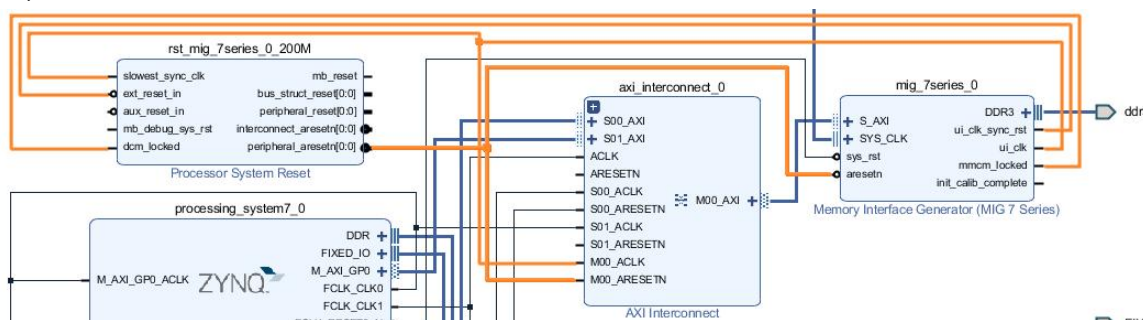
33) 点击 Run Connection Automation 自动连接



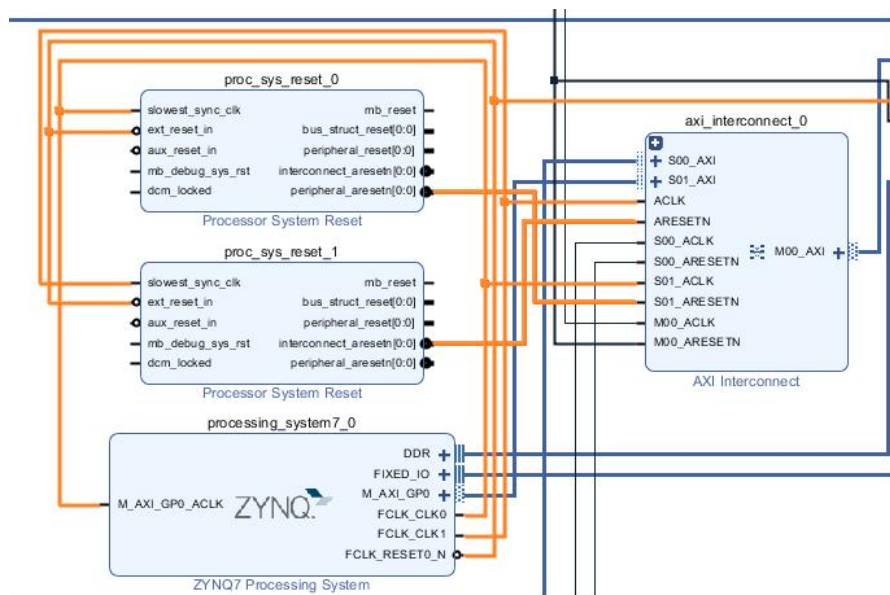
34) 点中 MIG 的 SYS_CLK 和 DDR3 端口右键选择 Make External，导出引脚



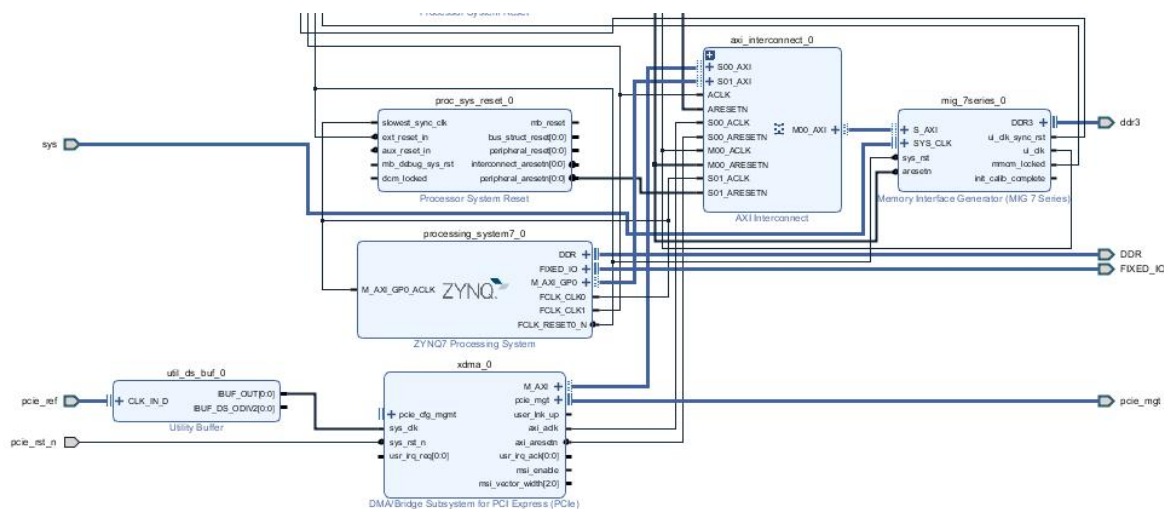
35) 添加 reset 模块，进行如下连接



36) 继续添加两个复位模块，分别连接 axi_interconnect 的复位 ARESETN 和 S01_ARESETN



37) 修改端口名称

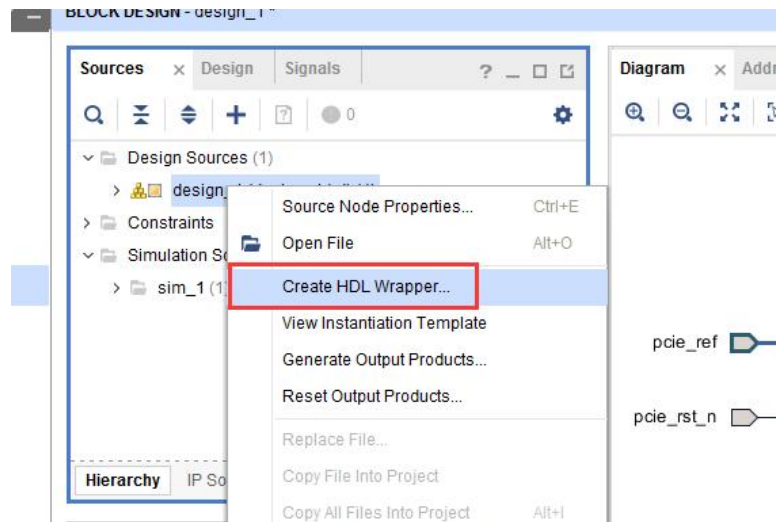


38) 查看 Address Editor, 点击自动分配地址, 由于 PCIe 上位机访问是从 0 地址开始的, 因此需要将 xdma 的 Offset Address 改成 0x0000_0000_0000_0000

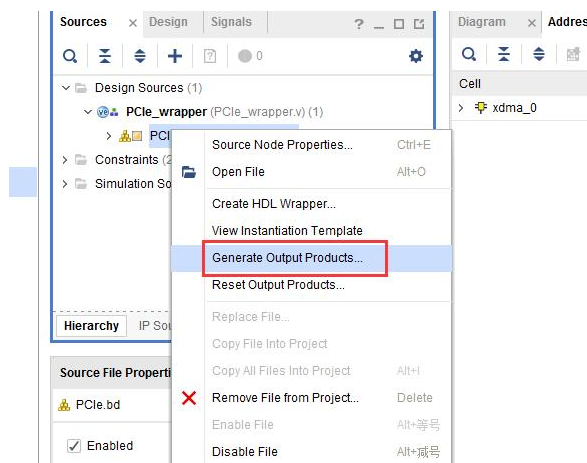
Cell	Slave Interface	Base Name	Offset Address	Range	High Address
processing_system7_0					
Data (32 address bits : 0x40000000 [1G])					
mig_7series_0	S_AXI	memaddr	0x4000_0000	1G	0x7FFF_FFFF
xdma_0					
M_AXI (64 address bits : 16E)					
mig_7series_0	S_AXI	memaddr	0x0000_0000_0000_0000	1G	0x0000_0000_0000_0000

39) 保存设计, 按 F6 检查设计

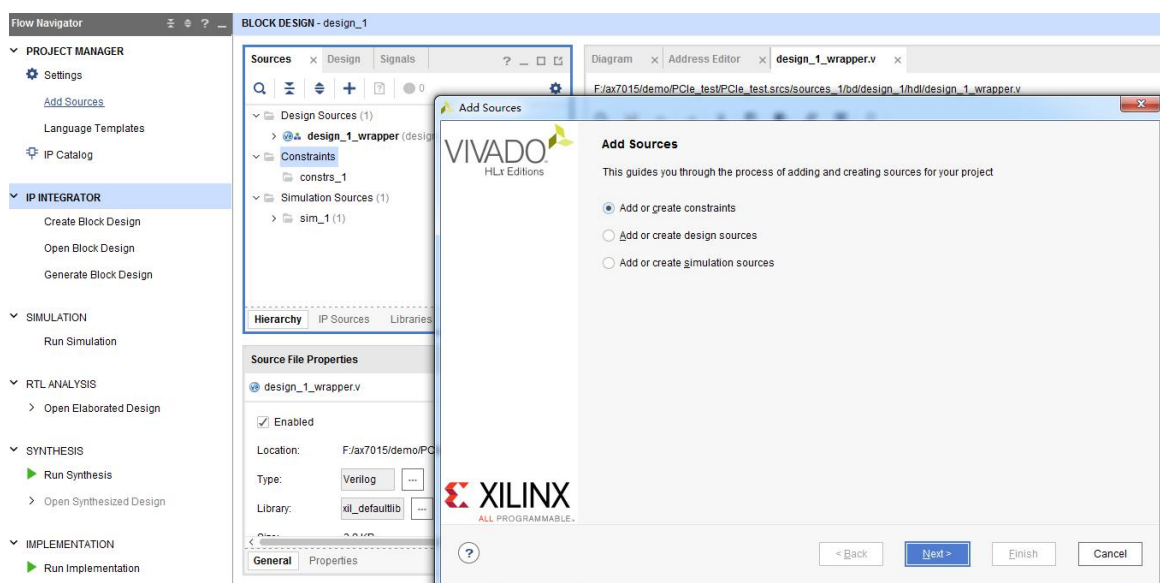
40) 创建 HDL 文件



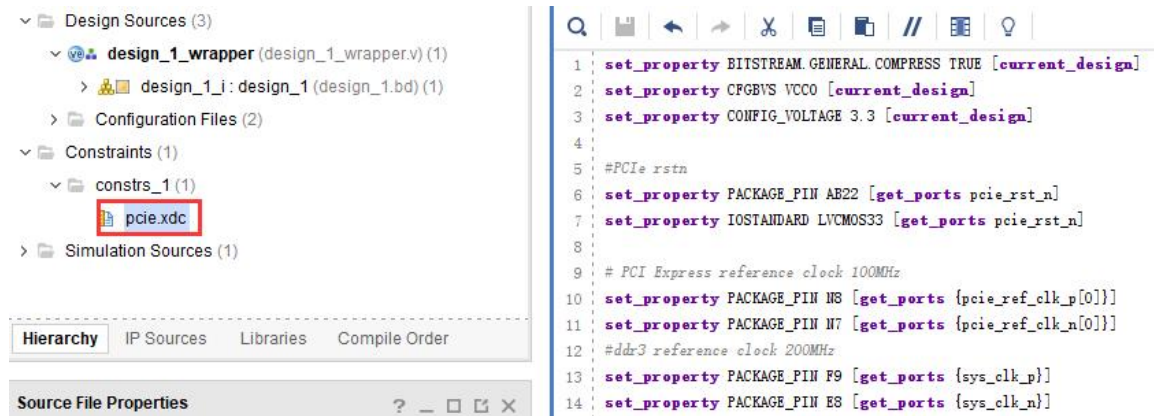
41) 运行 Generate Output Products



42) 添加 xdc 约束



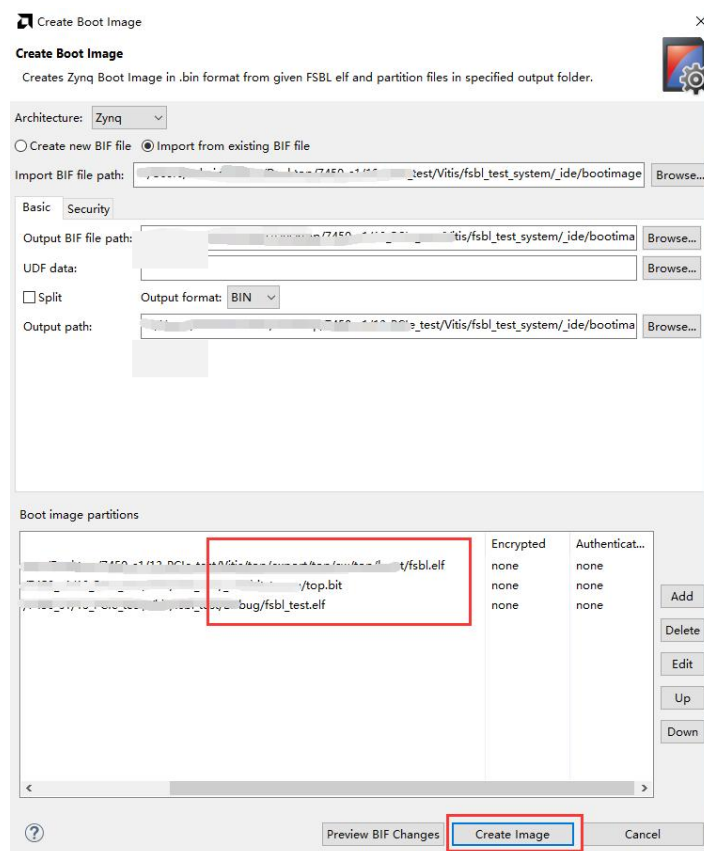
- 43) 修改 xdc 文件内容, 详细文件可以参考例程提供的 Vivado 工程, xdc 文件使用是 TCL 教程语言, 不熟悉 TCL 可以去找相关资料



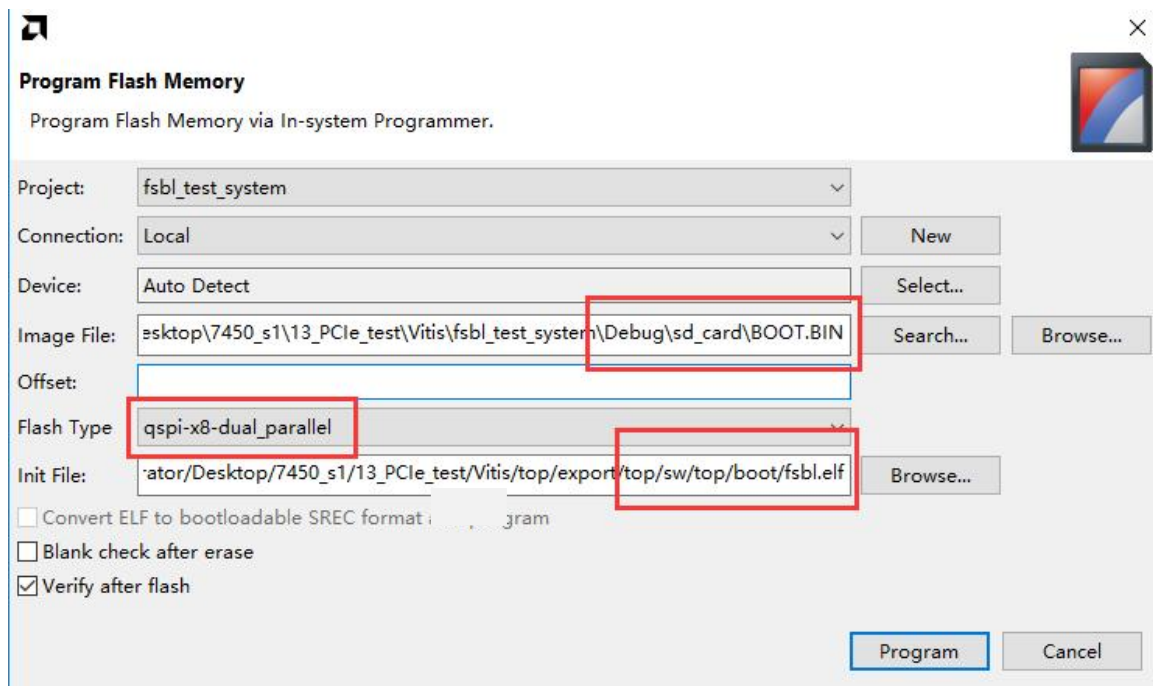
- #### 44) 编译生成 bit 文件

34.2 生成烧写 BOOT

- 1) 导出硬件，运行 Vitis
- 2) 新建名为 fsbl 的 APP，模板选择 Zynq FSBL
- 3) 生成 BOOT.bin



- 4) 烧写 BOOT.bin 到 QSPI flash, PCIe 对启动时间有严格要求,使用 QSPI 启动比 SD 卡启动要快。



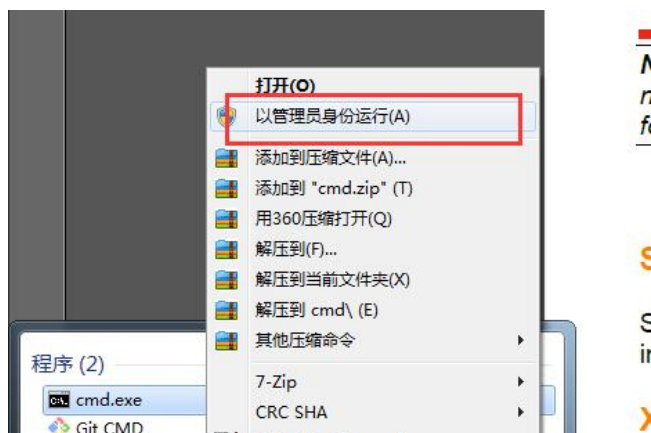
- 5) 烧写完成以后设置开发板启动模式为 QSPI，插入计算机 PCIe 插槽（断电操作），这时开发板不需要外部电源适配器供电，由电脑主板供电。

34.3 设置电脑进入测试模式

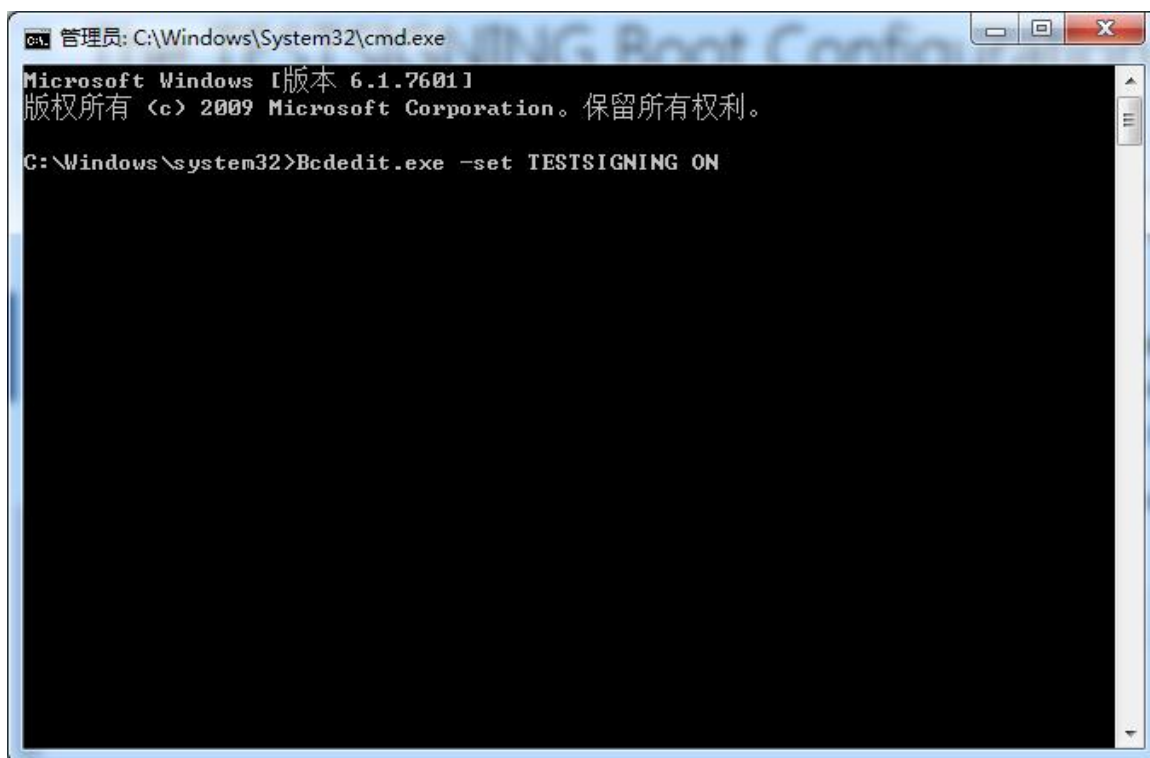
由于 PCIe 驱动程序没有经过微软数字签名，所以只能用于测试，需要将系统设置到测试模式，设置的详细方法参考 MSDN 文档

<https://msdn.microsoft.com/en-us/windows/hardware/drivers/install/the-testsigning-boot-configuration-option>

- 1) 使用管理员身份运行 cmd



- 2) 输入命令 Bcdedit.exe -set TESTSIGNING ON 打开测试模式



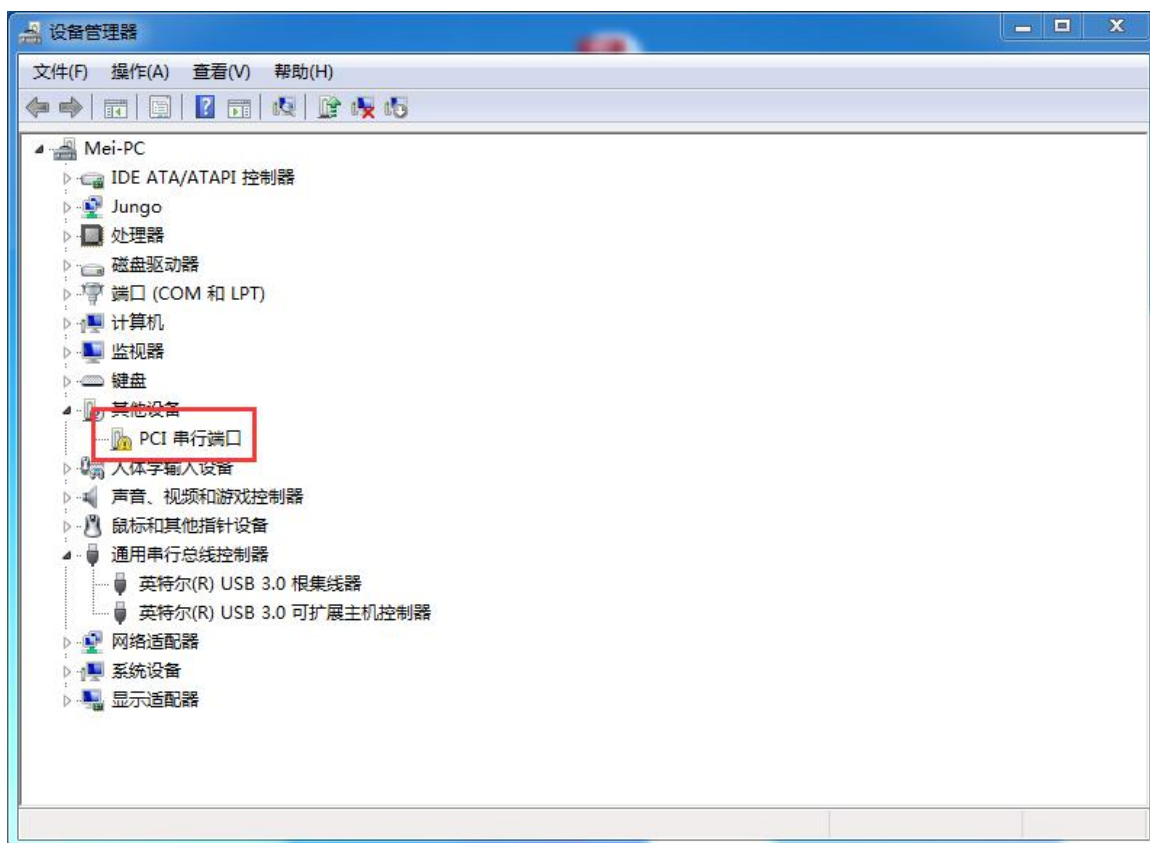
- 3) 重启电脑后桌面显示正在运行测试模式



34.4 安装 PCIe 驱动

PCIe 驱动和上位机测试文件都位于 “course_s2_Vitis/PCIe”

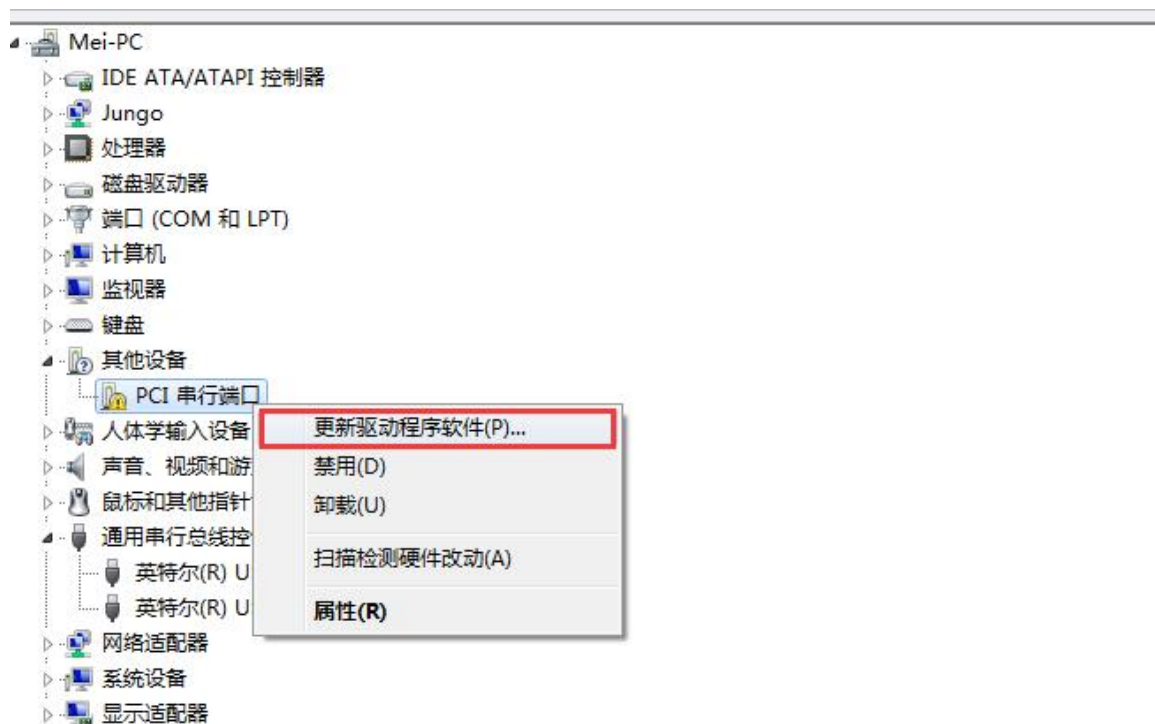
- 1) 没有安装驱动时，设备管理器如下，发现一个 PCI 串行端口



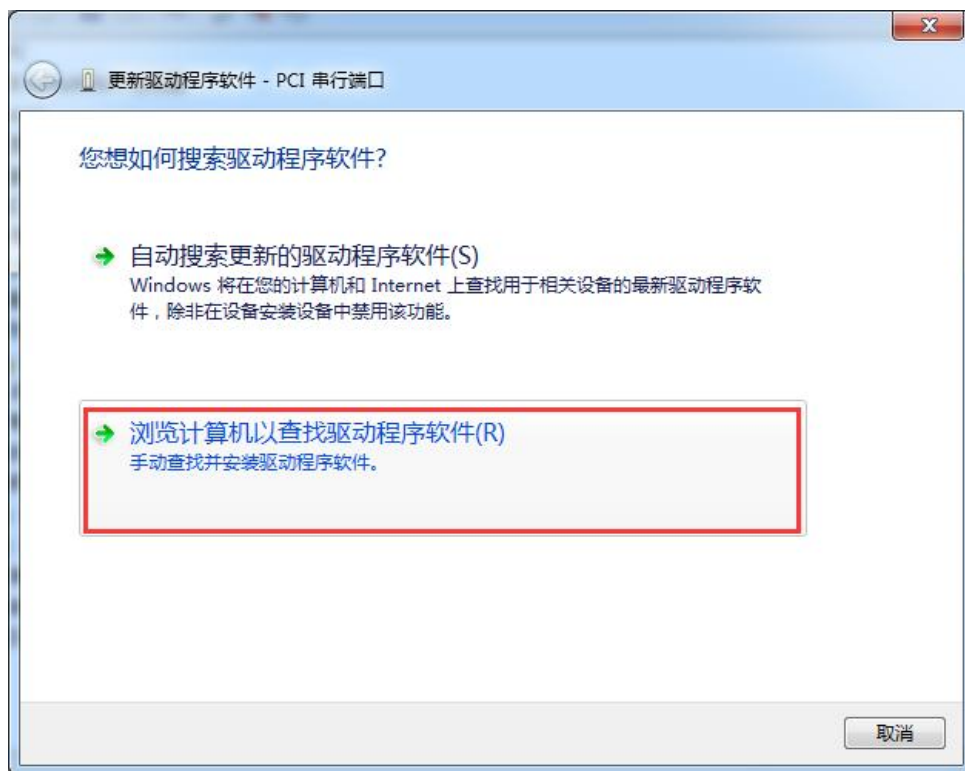
2) 例程提供的编译好的驱动程序



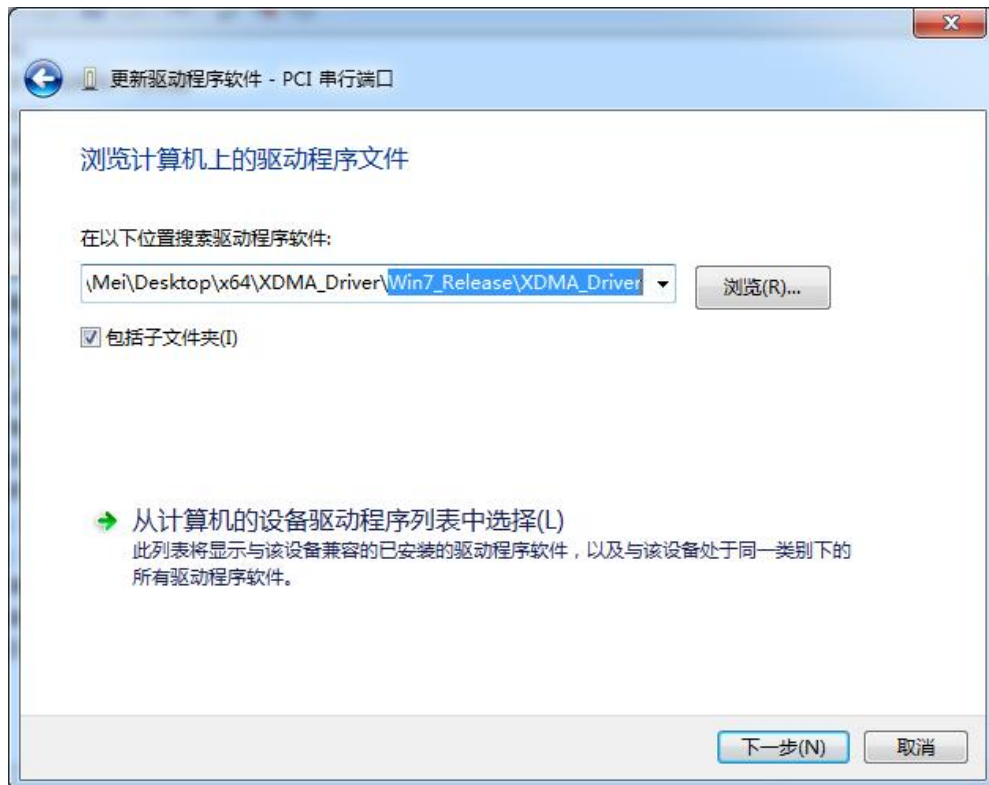
3) 选择设备，右键更新驱动



4) 浏览计算机以查找驱动程序软件



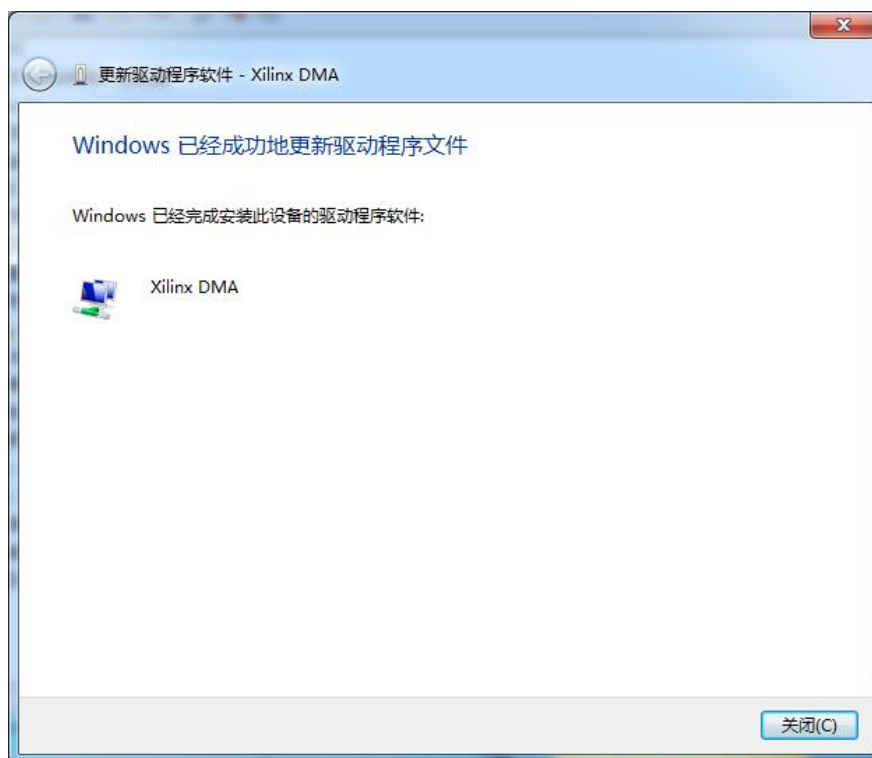
5) 测试电脑安装的是 win7，这里使用 Win7_Release 版本



6) 有个安全警告，选择安装

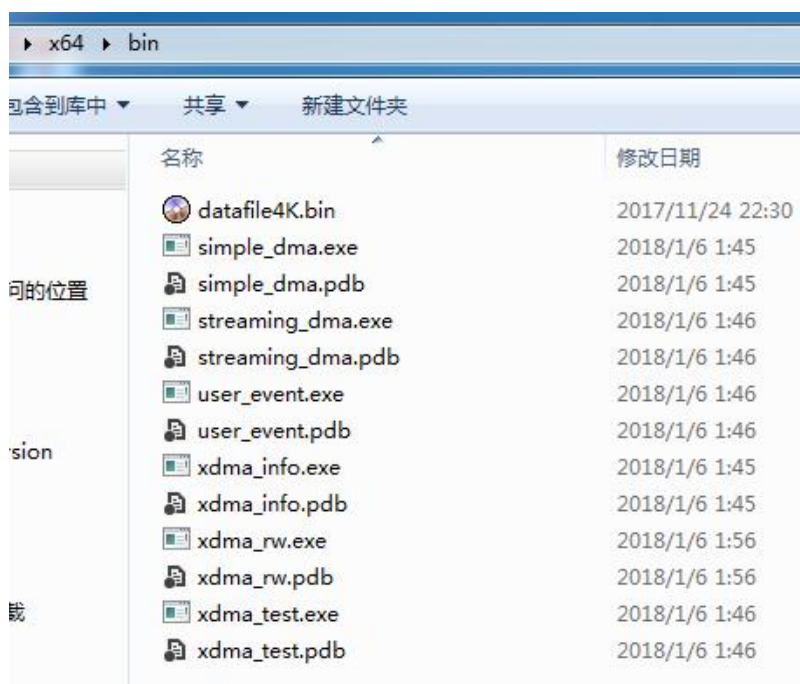


7) 正确安装了设备驱动



34.5 测试 PCIe

- 1) xilinx 提供了一些测试程序，不过都是命令行的程序



- 2) 芯驿电子开发了一些有界面的应用，使用 pciespeed.exe 来测试一下 PCIe 读写速率，这个读写测试会把数据写入 ZYNQ 的 ddr 然后再读取出来。

3) 读写同时进行测试



4) 只读测试



5) 只写测试



34.6 实验总结

本实验主要是掌握 PCIe 硬件工程的建立，初步体验 PCIe 读写速度，在后续的实验中我们会做一些更加实用的功能。上位机软件使用 QT 开发，如果需要自己修改编译，请先学习如何使用 QT。